Thesis

Physical Processes in the Interstellar Medium and the Formation of Molecular Clouds

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ABSTRACT

This thesis is devoted to the theoretical studies on the formation of molecular cloud. Understanding the formation of molecular clouds has a great importance in the theory of star formation and evolution of interstellar medium, because initial and boundary conditions of gravitational fragmentation that leads to star formation are determined in the formation epoch of molecular cloud. The study of the cloud formation is also important in understanding the phase change, circulation, and evolution of gasous matters in the Galaxy.

First, we re-analyse the the thermal and chemical equilibrium states of the interstellar medium of which density varies from 10^{-2} cm^{-3} to 10^{6} cm^{-3}. We have considered the following physical and chemical processes: (1) photoelectric heating from small grains and PAHs, (2) heating and ionization by cosmic rays and X-rays, (3) heating by H_2 formation and destruction, (4) atomic line cooling from hydrogen Lyα, C II, O I, Fe II, and Si II, (5) rovibrational line cooling from H_2 and CO, and (6) atomic and molecular collisions with grains. Results of this equilibrium calculation establish two stable branches in the phase diagram, and show basic properties of thermal and chemical equilibrium states of interstellar two-phase medium. The detailed heating and cooling functions of the interstellar medium and time dependent equations of chemical reactions provide useful basic tools for further analyses of dynamical problems.

Second, we propose that strong shock waves (e.g., galactic spiral shocks and supernova remnants) into an interstellar medium is one of the important mechanisms to produce dense clouds and molecular clouds. We have investigated the propagation of a shock wave into a warm neutral medium and a cold neutral medium by one-dimensional hydrodynamic calculations with detailed treatment of thermal and chemical processes. Our main result shows that thermal instability inside the shock-compressed layer produces a geometrically thin, dense layer in which a large amount of hydrogen molecules are formed. Linear stability analysis suggests that this thermally collapsed layer will fragment into small molecular cloudlets.

Third, we present results from the first high resolution two-dimensional calculations ever to follow the fragmentation due to thermal instability in shock-compressed layer. We use realistic thermal conduction coefficient to resolve "Field length" that is the critical wavelength for thermal instability. We find that geometrically thin cooling layer behind the shock front fragments into small cloudlets. The cloudlets have supersonic velocity dispersion in the warm neutral medium in which the fragments are embedded as cold condensations. The fragments tend to coalesce and become larger clouds.

Finaly, we investigate the evolution of cold clumps in warm medium in contrasting the difference with the previous semi-analytic works that include classical works of Zel'dovich & Pikel'ner (1969) and Penston & Brown (1970). Our main result shows that the true equilibrium of cold clouds and intercloud warm medium at the saturation pressure of the interstellar medium is attained within ~ 80 times the sound-crossing timescale in the intercloud warm medium, which is much shorter than the timescale previously believed.

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1

INTRODUCTION

Understanding processes leading to formation of stars is one of fundamental challenges in astronomy and astrophysics. In order to fully understand the formation of stars, one must study both the prenatal material and the recent products of star formation. Over the last two decades, radio and infrared observations have revealed that stars in our Galaxy form from the densest regions of molecular clouds. It is therfore important for star formation studies to acquire a knowledge of distribution and properties of dense gas within molecular clouds. However, the problem of how interstellar molecular clouds are formed has not yet been solved both in theoretically and observationally. This motivates us to do this work.

In this chapter, we present historical review on theoretical and observational backgrounds behinds this work in order to clarify the problems to be solved. In §1.3 purpose of this work is summarized.

1.1 Star Formation

1.1.1 Role of Turbulence

The interstellar medium (ISM) and cold gas clouds are characterized by a clumpy substructure and a turbulent velocity field (Larson 1981). The maintenance and dissipation processes of the turbulence are supposed to be important in the theory of star formation (McKee 1989, Nakano 1998, Elmegreen 1999). The understanding of the origin of cold clouds and their internal substructure has therefore fundamental importance for a consistent theory of star formation and ISM.

Some of the polarization maps and direct measurements of field strength in some star forming regions have suggested the importance of MHD (Alfvén waves) turbulence (Myers & Goodman 1988, Crutcher et al. 1993). Recent simulations of MHD turbulence, however, suggest that it dissipates rapidly (Gammie & Ostriker 1996, Ostriker, Gammie, & Stone 1999). Possible sources of energy supply are winds and outflows from young stellar objects (Franco & Cox 1983, McKee 1989). Note however that clumpy structures with supersonic velocity dispersions are also observed even in regions where star formation is inactive. Thus, the origin of clumpy cloud structure remains as an outstanding issue.

1.1.2 Initial Condition of Gravitational Fragmentation

In order to understand the stellar initial mass function (IMF), many authors have studied the origin of the mass function of molecular cloud cores where stars form since the mass of the cloud core may have a close relation with that of a star. Indeed, in the Taurus dark cloud, the masses of newly formed stars and the masses of the associated dense cores in molecular clouds are of the same order(e.g., Myers & Benson 1983). However, detailed observations of the mass function of dense cores have only recently been initiated(Blitz 1993; Tatematsu et al. 1993). Inutsuka & Miyama (1992, 1997) are devoted to the theoretical study of collapse and fragmentation processes within molecular clouds. Using their results, Inutsuka (2000; see also Inutsuka & Tsuribe 2000) produced a theory for the expected mass function of dense molecular cloud cores. He have suggested that the IMF of the cloud core may depend on the initial condition (e.g., density and velocity fluctuation spectrum) of the molecular clouds. Therefore the studies of the formation of molecular clouds have great importance in the theory of star formation because it provides the initial condition of gravitational fragmentation.

1.2 Models of the Interstellar Medium

Models of ISM have been studied by many authors. Field, Goldsmith, & Habing(1969, hereafter FGH) first showed that a cold ($T \sim 50$ K) neutral medium (CNM) and a warm ($T \sim 8000$ K) neutral medium (WNM) can be in pressure equilibrium. Cox & Smith (1974) showed that supernova explosions can greatly modify the general aspect of the ISM. McKee & Ostriker (1977) developed a theory of a supernova-dominated ISM. The late phase supernova remnant (SNR) expands until the internal pressure drops to the ambient interstellar pressure ($n = 0.1$ cm^{-3}, $T = 10^4$ K). This maximum radius $R_{\rm max} = 10^{2.1}$ pc is reached at a time $t_{\rm max} = 10^{6.3}$ yr. The total volume $SR_{\rm max}^3 t_{\rm max}$ which all SNRs occupy in the maximum expansion time $t_{\rm max}$ exceeds the volume of the Galactic disk, where $S \sim 10^{-2}$ yr^{-1} is the supernova rate in our Galaxy. This means that the SNRs overlap before they are dissipated. Thus, the study of shock propagation into the ISM has fundamental importance of the theory of the ISM.

1.2.1 H I Clouds and Molecular Clouds

The tiny scale structures on scales of tens of astronomical units were observed in the ISM. These structures have been detected by 21 cm absorption lines against quasars with VLBI techniques (Dieter, Welch, & Romney 1976), against pulsars with time variability (Frail et al. 1994) and against close binary stars in optical interstellar lines (Meyer & Blades 1996). The tiny scale structure is seen in all directions for which it has been searched, which suggests that it is ubiquitous, not associated with large extinction (Heiles 1997). AU-sized structures have been seen not only in atomic gas but also in molecular gas. Langer et al. (1995) observed

several clumps ranging in size from 0.007 to 0.021 pc in the Taurus molecular cloud 1 with interferometer techniques. The mass of these smallest fragments is of the order of $0.01 M_{\odot}$. These small-scale structures appear to be gravitationally unbound by a large factor. The presence of these tiny scale atomic and molecular structures in a variety of the ISM suggests that their formation mechanisms have some similarities.

1.2.2 Role of Thermal Instability

We propose that the clumpiness in clouds arises naturally from their formation through thermal instability which acts on time scales that can be much shorter than the duration of the Interstellar shocks (e.g., galactic spiral shocks and supernova remnants). The basic properties of the thermal instability are studied in a pioneering paper by Field (1965). Schwarz, McCray, & Stein (1972) studied numerically the growth of condensation in cooling region including the effects of ionization and recombination. Hennebelle & Pérault (1999) studied the elementary condensation process in turbulent flow in the restricted conditions of neutral atomic gas in plane-parallel geometry. Burkert & Lin (2000) studied cooling and fragmentation of gas using simplified power-law cooling function. Smith (1980) studied collisions between cold atomic clouds, which produce thick layers of shock-heated atomic gas in which thin sheets of cold molecular gas form by thermal instabilities. Koyama & Inutsuka (2000, hereafter Paper I) have done one-dimensional hydrodynamic calculations for the propagation of a strong shock wave into warm neutral medium (WNM) and cold neutral medium (CNM) including detailed thermal and chemical processes. They have shown that the postshock region collapses into a cold layer as a result of thermal instability. They expect that this layer will break up into very small cloudlets which have different translational velocities.

1.2.3 Characteristics of Pressure-Confined Clouds

Interstellar clouds, even though not gravitationally bound, can exist in pressure equilibrium with interstellar medium; their internal pressure is balanced by the pressure of an external diffuse medium. In an important paper, FGH presented a model of the interstellar clouds which explained many of the observed properties of the interstellar medium. Under the influence of cosimc ray heating and cooling by collisional excitation and subsequent emission of a variety of spectral lines from different atoms and ions, the interstellar medium comes to an equilibrium temperature, which is a function of the gas density and increases with decrese in density. Subsequent analysis, however, indicated that the cosmic ray ionization rate is significantly lower (Spitzer & Jenkins 1975). Another potential heating source is the photoelectric ejection of electrons from dust grains by the interstellar radiation field (Watson 1972; Glassgold & Langer 1974; Jura 1976; Draine 1978; de Jong 1977, 1980). d'Hendecourt & Léger (1987) and Lepp & Dalgarno (1988) demonstrated that photoelectric emission from very small grains and large molecules (polycyclic aromatic hydrocarbons, or

PAHs) could potentially be an important heating source. Recently, Bakes & Tielens (1994) have calculated the photoelectric heating rate for a size distribution of particles [$n(a)da \propto a^{-3.5}da$] extending from 100 to 3 Å in radius. Using their results, substantial refinements of the similar properties of the neutral atomic phases of the interstellar medium have been proposed by Wolfire et al. (1995). If the pressure in these equilibrium states is plotted against density, it is found that there is a range of pressures for which three density states exist corresponding to a single pressure. Of these states one is thermally ustable and FGH identified the other two with the interstellar clouds and the intercloud medium which can thus exist together in pressure equilibrium.

Pressure equilibrium is not the only necessary condition for the equilibrium of an interstellar cloud. The interface between cloud and intercloud medium as a whole must be in thermal balance as heat diffuses from regions of the interface that are heated excessively to those that would otherwise cool. Zel'dovich & Pikel'ner (1969, hereafter ZP) and Penston & Brown (1970, hereafter PB) have found quite independently that it is only at the saturated vapour pressure that the clouds and intercloud medium are in true equilibrium. If clouds and intercloud medium are not in true equilibrium, cloud enter into a phase of stationary growth (i.e., evaporate or condense) driven by thermal conductivity. The stationary propagation of thermal fronts has been repeatedly considered since ZP and PB. They analyzed in detail the structure of the cloud-intercloud interface for a plane-parallel one-dimensional steady cooling flow model, determining the evaporation or condensation rates as a function of pressure. They have found that the timescale for evaporation or condensation of one solar mass was $\sim 10^9 - 10^{10}$ years, and they concluded that evaporation or condensation could be neglected because the timescale is much longer than those for other dynamical phenomena, such as the cloud-cloud collision time $\sim 10^7$ years, the sound travel time through a cloud ($\sim 10^7$ years)(Shu et al. 1972), the galactic rotation time ($\sim 10^8$ years), and the cloud foramtion time ($\sim 10^7$ years)(Goldsmith 1970). Elphick, Regev, & Shaviv (1992) included flow equations, but restricted the study to isobaric conditions. Ferrara & Shchekinov (1993) avoided this restriction considering quasi-isobaric solution. Hennebelle & Pérault (1999) recalled the derivation of the front velocity in the quasi-isobaric limit. Graham & Langer (1973) and Parravano (1987) studied the one-dimensional model in spherical-symmetric geometry. In spite of these developments, none of the previous works solved dynamical equation of motion, and the resultant evolution timescale they obtained are similar. their timescale is similar order of magnitude within the hydrostatic approach.

On the other hand, hydrodynamical effect becomes important even if cloud and intercloud have almost the same pressure. Because pressure irregularites are removed rapidly on the sound-crossing timescale via hydrodynamical effects. Therefore it is imperative to study the fully nonlinear hydrodynamical calculation to analyze the evolution of two phase system.

1.3 Purpose of This Work

Main goals of this thesis are summarized as follows:

1. Physical processes that include the radiative heating and cooling, and chemical reactions of atoms and molecules are analysed by extending the previous studies of neutral atomic phases.

2. Linear stability analysis of thermal instability in isobarically contracting gas provides basic properties of fragmentation of shock-compressed layers.

3. From the numerical simulations of shock-compressed layer, we demonstrate the fragmentation of thermally collapsed layer and the role of thermal instability behind the shock front.

4. We investigate the evolution of cold clumps in warm medium in contrasting the difference with the previous semi-analytic works that include classical works of Zel'dovich & Pikel'ner (1969) and Penston & Brown (1970).

In this thesis we first analyse the thermal and chemical processes that are incorporated into the subsequent numerical calculations in Chapter 2. We perform the linear stability analysis of thermal instability in isobarically contracting medium in Chapter 3. In Chapter 4, we investigate the propagation of shock wave into WNM and CNM by one-dimensional hydrodynamic calculations with detailed treatment of thermal and chemical processes described in chapter 2. Chapter 5 is one of the main results in the thesis, where the first high resolution two-dimensional hydrodynamic calculations ever to follow the fragmentation due to thermal instability in shock-compressed layer are presented. In Chapter 6, we investigate fundamental processes in the dynamical evolution of interstellar two phase medium by one-dimensional hydrodynamic calculations with thermal conductivity. Finally we summarize our conclusion of this thesis in Chapter 7.

2

THERMAL AND CHEMICAL PROCESSES IN INTERSTELLAR MEDIUM

In this Chapter, thermal and chemical processes in our calculation are summarized. We included the formation and cooling of H_2 and CO in addition to the processes described by Wolfire et al. (1995, hereafter W95).

2.1 Heating and Cooling Processes

2.1.1 Photoelectric Heating from Small Grains and PAHs

The photoelectric emission from small grains and polycyclic aromatic hydrocarbons (PAHs) induced by FUV photons is an important mechanism for heating a diffuse interstellar gas cloud. We use the results of Bakes & Tielens (1994, hereafter BT) to calculate the photoelectric heating. In their model, grains are distributed with a Mathis, Rumpl, & Nordsieck (1977, hereafter MRN) power-law distribution in size, $n(a)da \propto a^{-3.5}da$, between 3 and 100 Å in radius. BT calculated the charge distribution for each particle size taking the photoionization and the electron and positive ion recombination into account and found the total photoelectric heating from the distribution of particles. The resulting heating rate is given by

$$\Gamma_{\rm pe} = 1.0 \times 10^{-24} \epsilon G_0 \,{\rm ergs\,s^{-1}}, \tag{2.1}$$

where ϵ is the heating efficiency and G_0 is the incident FUV field normalized to the local interstellar value ($= 1.6 \times 10^{-3} {\rm ergs\,cm^{-2}s^{-1}}$) estimated by Habing (1968). BT provide a simple fit to ϵ as a function of $G_0 T^{1/2}/n_{\rm e}$:

$$\epsilon = \frac{4.9 \times 10^{-2}}{1.0 + [(G_0 T^{1/2}/n_{\rm e})/1925]^{0.73}} + \frac{3.7 \times 10^{-2} (T/10^4)^{0.7}}{1.0 + [(G_0 T^{1/2}/n_{\rm e})/5000]}, \tag{2.2}$$

where $n_{\rm e}$ is the electron density.

Grains and PAHs may also be an important gas coolant. We treat the heating caused by the electron ejection (eq.[2.1]) and the recombination cooling separately, so that the net

heating or cooling is the difference between the heating and cooling. We use the fit to the recombination cooling provided by BT:

$$\Lambda_{\rm pe} = 4.65 \times 10^{-30} T^{0.94} (G_0 T^{1/2}/n_{\rm e})^{\beta} n_{\rm e} \,{\rm ergs\ s^{-1}}, \tag{2.3}$$

with $\beta = 0.74/T^{0.068}$.

2.1.2 Ionization Heating by Cosmic Rays and Soft X-Rays

When the UV flux is strongly attenuated by hydrogen photoabsorption, cosmic rays provide a main heating mechanism. We adopt a primary cosmic-ray ionization rate of $\zeta_{\rm CR} = 1.8 \times 10^{-17}{\rm s}^{-1}$ (including the ionization of He) following W95. Primary electrons ejected by cosmic-ray ionization have a larger kinetic energy than that of thermal electrons. This primary electron heats surrounding gas by secondary ionization. The total heating rate is given by

$$\Gamma_{\rm CR} = \zeta_{\rm CR} E_h(E, x_{\rm e}), \tag{2.4}$$

where the function $E_h(E, x_{\rm e})$ gives the heat deposited for each primary electron of energy E (Shull & Steenberg 1985).

Soft X-rays also ionize and heat the ISM. W95 calculated the heating and ionization rates using the observed diffuse X-ray spectrum and photoionization cross section of H, He, and other trace elements. The analytic fits are given in their paper. Soft X-ray is attenuated by photoabsorption of neutral hydrogen atom. We use the value of absorbing column density 10^{19} –10^{20} cm^{-3} as a standard H I clouds.

2.1.3 Formation and Photodissociation Heating from H_2

When a hydrogen molecule is photodissociated, the chemical potential of the gas is raised by the dissociation energy, 4.48 eV. In addition to this, a small amount of kinetic energy is deposited to the photodissociated atoms. According to Hollenbach & McKee (1979, hereafter HM79), the heating rate is

$$\Gamma_{\rm UV} = 9 R_{\rm pump} \left\{ (2.2{\rm eV}) \left[1 + n_{\rm cr}(H_2)/n \right]^{-1} \right\}, \tag{2.5}$$

where

$$n_{\rm cr} = \frac{10^6 T^{-1/2}}{1.6 x_{\rm H} \exp[-(400/T)^2] + 1.4 x_2 \exp\{-[12000/(T+1200)]\}} \ \ {\rm cm^{-3}}, \tag{2.6}$$

and the rate coefficient $R_{\rm pump}$ is given in §2.2.2.

When a hydrogen molecule forms through associative detachment or catalytic reaction, it is in a highly excited rovibrational state. Only a small fraction of the formation energy is

available for heating the gas, while most of the energy escapes the cloud via line emission. We use the heating rates from HM79 as follows:

$$\Gamma_{\rm gr} = R x_{\rm H} \left\{ 0.2 + 4.2[1 + n_{\rm cr}({\rm H}_2)/n]^{-1} \right\}, \tag{2.7}$$

and the rate coefficient R is given in chapter 2.2.2.

2.1.4 Atomic Line Cooling

At the high temperatures ($T \gtrsim 8000$ K) collisional excitation of hydrogen Lyα can contribute to the cooling. We also include low-lying metastable transitions of C II (2326Å), O I (6300Å), Fe II (5.3 μm), and Si II (2240Å), using rates from Hollenbach & McKee (1989, hereafter HM89).

Collisional excitation of the fine-structure lines of C II (158 μm) and O I (63 μm) is a dominant gas coolant at temperatures $T \lesssim 8000$ K. Additional cooling is provided by the fine-structure transitions of the ground electronic state of Fe II (26 μm), Si II (35 μm) (HM89).

We adopt trace element abundance following W95; $x_{\rm He} = 0.1$, $x_{\rm O} = 4.6\times10^{-4}$, $x_{\rm C} = 3\times10^{-4}$, $x_{\rm Si} = 3.55\times10^{-6}$, and $x_{\rm Fe} = 7.08\times10^{-7}$. Several trace elements, C, Si, and Fe, have ionization potentials less than 13.6 eV and thus are susceptible to ionization by the interstellar UV field. For convenience, the level of ionization of these trace elements is assumed to be in singly charged. On the other hand, oxygen remains neutral by charge exchange reaction to hydrogen, because oxygen has almost the same ionization potential as hydrogen's.

2.1.5 H_2 Cooling

At temperatures above 500 K, rovibrational lines of H_2 contribute to the cooling function. Following HM79 we adopt

$$\Lambda_{\rm H_2} \approx n({\rm H}_2) \left[x_{\rm H} \frac{L_{\rm vr}^{\rm H}({\rm LTE})}{1 + n_{\rm cr}^{\rm H}/n} + x_2 \frac{L_{\rm vr}^{\rm H_2}({\rm LTE})}{1 + n_{\rm cr}^{\rm H_2}/n} \right], \tag{2.8}$$

where $n_{\rm cr}$ is the critical density defined as follows:

$$n_{\rm cr}^{\rm H,H_2} \equiv \frac{L_{\rm vr}^{\rm H,H_2}({\rm LTE})}{L_{\rm vr}^{\rm H,H_2}(n \to 0)} n, \tag{2.9}$$

which depends on temperature only, not on density. We use the function $L_{\rm vr}^{\rm H,H_2}({\rm LTE})$, $L_{\rm vr}^{\rm H,H_2}(n \to 0)$ from HM79. We update the function $L_{\rm r}^{\rm H}(n \to 0)$ from Galli & Palla (1998). We assume an ortho – H_2 to para – H_2 ratio of 3:1.

2.1.6 CO Cooling

Rotational line emission from CO is a major low-temperature cooling agent for molecular clouds. The CO rotational cooling is derived from McKee et al. (1982). The optically thin cooling function is

$$\Lambda_{\rm CO}({\rm rot}) = \frac{4(k_{\rm B}T)^2 A_0}{E_0[1+(n_{\rm cr}/n)+1.5(n_{\rm cr}/n)^{1/2}]}, \tag{2.10}$$

where $A_0 = 9.7\times10^{-8}{\rm s}^{-1}$, $E_0/k_{\rm B} = 2.76$ K , $n_{\rm cr} = 3.3\times10^6 T_3^{3/4}\,{\rm cm}^{-3}$ for ^{12}CO (McKee et al. 1982).

The CO vibrational cooling at the lowest level is

$$\Lambda_{\rm CO}({\rm vib}) = \sum_{\rm H,H_2} \gamma_{01}^{\rm H,H_2}\Delta E_{10}, \tag{2.11}$$

where $\Delta E_{10} = 3080{\rm K}/k_{\rm B}$. The vibrational rate coefficients for transitions to the v=1 state of CO were taken from HM89.

2.1.7 Atomic and Molecular Collisions with Dust Grains

The cooling of the gas by cooler dust grains is important at high densities $n \gtrsim 10^5\,{\rm cm}^{-3}$. The cooling rate per hydrogen nuclei is

$$\Lambda_{\rm gr} = 1.2\times10^{-31} n \left(\frac{T}{1000{\rm K}}\right)^{1/2}\left(\frac{100\,{\rm \AA}}{a}\right)^{1/2}\times[1-0.8\exp(-75/T)](T-T_{\rm gr})\ {\rm ergs\ s}^{-1}, \tag{2.12}$$

where $T_{\rm gr}$ is an effective grain temperature, averaged over the assumed MRN size distribution (HM89). Since the gas generally collides with the smaller grains, we take $T_{\rm gr}$ to be the temperature 8 K of grains of radius $a = 100\,{\rm \AA}$.

2.2 Chemical Reactions

For the range of temperatures and densities considered, the recombination time scale is longer than the cooling timescale (See Figure 4.1d). The residual electrons make the cooling rates larger than that of equilibrium abundance. Therefore, it is imparative to treat chemical non-equilibrium processes.

The main coolant in dense clouds is supposed to be CO, so that CO formation should be taken into account. Although H_2 is minor coolant, H_2 enables the formation of CO. Thus, we should take into account the formation of H_2.

2.2.1 Ionization and Recombination of Hydrogen Atom

Photoionizing UV fields are attenuated by large absorption cross section of hydrogen atom. The hydrogen atom is also ionized by cosmic rays and X-rays. The total ionization rates per hydrogen atom are given in the Appendix of W95. Collisional ionization is also important in shock-compressed layers, and rate coefficient of this processes is $k_{\rm I} = 5.8 \times 10^{-9} T_4^{0.50} \exp(-15.8/T_4)\,{\rm cm^{-3}s^{-1}}$(HM79). The radiative recombination coefficient of hydrogen is given by Shapiro & Kang (1987).

2.2.2 Formation and Dissociation of H_2

H_2 is formed catalytically on dust grains and by associative detachment; it is destroyed by collisional dissociation, by photodissociation via the Lyman-Werner electronic transitions, and by cosmic rays.

Following Tielens & Hollenbach (1985, hereafter TH85), the rate at which H_2 is produced catalytically on dust grains is

$$R = 0.5\bar{v} n_{\rm H} n_d \sigma_d S(T) \approx 6 \times 10^{-17} (T/300)^{0.5} n_{\rm H} n S(T)\,{\rm cm^{-3}s^{-1}}, \tag{2.13}$$

$$S(T) = [1 + 0.04(T + T_{\rm gr})^{0.5} + 2 \times 10^{-3} T + 8 \times 10^{-6} T^2]^{-1}, \tag{2.14}$$

where $\bar{v}$ is the mean thermal velocity of the atomic hydrogen, n_d is the number density of dust grains, and σ_d is the average dust grain cross section.

H_2 can be formed by associative detachment of the H^- ion,

$$H + H^- \rightarrow H_2 + e^-. \tag{2.15}$$

We use the formation rates from HM79 to obtain an effective rate coefficient.

The shock waves can produce sufficiently high temperatures so that collisional dissociation of H_2 should be considered. The rate of dissociation of H_2 by H atoms is $k_{\rm D} = 3.4 \times 10^{-9} \exp(8000/T) \exp(-5.19 \times 10^4/T)\,{\rm cm^{-3}s^{-1}}$ (HM79). The rate of dissociation of H_2 by electron impact with LTE vibrational population is taken into account (Stibbe & Tennyson 1999). Although these LTE reaction rates are overestimates for H_2 dissociation, there is no significant difference between models with and without collisional dissociation. H_2 forms after the gas layer collapses because the timescale of H_2 formation is much larger than cooling timescale. The temperature in the collapsed layer is too low for H_2 to dissociate by collision. This is why collisional dissociation is ineffective in our models.

We adopt the rate of photodissociation from TH85,

$$R_{\rm pump} = 3.4 \times 10^{-10} G_0 \beta(\tau)\,{\rm s^{-1}}, \tag{2.16}$$

where $\beta(\tau)$ is the probability that photons of the interstellar radiation field penetrate to optical depth τ in a plane parallel cloud layer. Photodissociating UV fields are self-shielded by H_2, and the optical depth τ is

$$\tau = 1.2 \times 10^{-14} N(\mathrm{H_2}) \delta V_D^{-1}, \tag{2.17}$$

where $N(\mathrm{H_2})$ is the column density of $\mathrm{H_2}$ and δV_D is the Doppler line width in kilometers per second, and we adopt this value to be the sound velocity. The analytic expression of $\beta(\tau)$ is given by TH85.

The rate at which cosmic rays destroy $\mathrm{H_2}$ is estimated by HM89 to be $2.29\zeta_{\mathrm{CR}}$, where ζ_{CR} is the primary cosmic-ray ionization rate for atomic hydrogen.

2.2.3 Carbon Chemistry

We include a simplified treatment of the conversion of singly ionized carbon $\mathrm{C^+}$ to carbon monoxide CO following Langer (1976) and Nelson & Langer (1997). The simplified chemical model assumes the direct conversion of $\mathrm{C^+}$ to CO, or vice versa, without accounting explicitly for the intermediate reactions. The equation describing the rate of production and destruction of CO is written

$$\frac{dn(\mathrm{CO})}{dt} = k_0 n(\mathrm{C^+}) n\beta - \Gamma_{\mathrm{CO}} n(\mathrm{CO}) \tag{2.18}$$

where $k_0 = 5 \times 10^{-16}\,\mathrm{cm^3 s^{-1}}$, $\Gamma_{\mathrm{CO}} = 10^{-10} G_0\ \mathrm{s^{-1}}$, and β is defined by the expression

$$\beta = \frac{k_1 x(\mathrm{O\ I})}{\mathrm{k_1 x(O\ I) + G_0[\Gamma_{CHx}/n(H_2)]}}. \tag{2.19}$$

The definition of Γ_{CHx} is given by $\Gamma_{\mathrm{CHx}} = 5 \times 10^{-10} G_0$, and k_1 takes the value $k_1 = 5 \times 10^{-10}$. For CO photodissociation we do not treat the line self-shielding of CO following Nelson & Langer (1997), because we are only interested in the beginning of CO formation.

Species(0,1,2)[a]	E/k_B[K][b]	$\lambda[\mu m]$	$n^e_{\rm cr}({\rm cm}^{-3})$	$n^H_{\rm cr}({\rm cm}^{-3})$	$A_{ij}[s^{-1}]$	$\gamma^e_{ij}[cm^{-3}$
CII($^2P_{1/2}$, $^2P_{3/2}$)	9.2(1)	157.7	$8.7(0)T_2^{0.50}$	$3.0(2)T_2^{-0.07}$	2.4(-6)	$2.8(-7)T_2^-$
CII(^2P, ^4P)	6.2(4)	2326Å	$1.6(8)T_4^{0.50}$	3.6(12)	3.6	$2.3(-8)T_4^-$
FeII($^6D_{9/2}$, $^6D_{7/2}$, $^6D_{5/2}$)	5.6(2)	26.0	$1.2(3)T_2^{0.41}$	$2.2(6)T_2^{-0.09}$	2.1(-3)	$1.8(-6)T_2^-$
	9.6(2)	15.0	$6.0(2)T_2^{0.50}$	1.5(6)	1.5(-9)	$1.8(-6)T_2^-$
	4.1(2)	35.4			1.6(-3)	$8.7(-7)T_2^-$
FeII($^6D_{9/2}$, $^4F_{9/2}$, $^4F_{7/2}$)	2.7(3)	5.34	$1.0(4)T_4^{0.50}$	$1.6(8)T_4^{-0.11}$	2.8(-4)	$2.2(-8)T_4^-$
	3.5(3)	4.12	$1.4(5)T_4^{0.50}$	2.9(9)	1.0(-4)	$7.1(-9)T_4^-$
	8.1(2)	17.9			6.1(-3)	$3.8(-8)T_4^-$

3

THERMAL INSTABILITY IN ISOBARICALLY CONTRACTING GAS

To study the characteristic length and timescale of the thermally collapsed layer, we have done linear stability analysis. Field (1965) analyzed the stability of uniform gas in thermal equilibrium. Schwarz, McCray, & Stein (1972) analyzed the isochorically cooling gas without heating. Our non-linear calculation, however, shows that the shock-compressed layer is almost isobaric. In this appendix, we analyze the stability of isobarically contracting gas.

3.1 The Basic Equations

We consider the gas with number density n, temperature T, velocity v, and pressure P, where all variables are functions of position and time. The governing hydrodynamic equations are

$$\frac{dn}{dt} + n\frac{\partial v}{\partial X} = 0, \tag{3.1}$$

$$m_{\rm H} n \frac{dv}{dt} = -\frac{\partial P}{\partial X}, \tag{3.2}$$

$$\frac{1}{\gamma - 1}\frac{dP}{dt} - \frac{\gamma}{\gamma - 1}\frac{P}{n}\frac{dn}{dt} = n\Gamma - n\Lambda + \frac{\partial}{\partial X}\left(K\frac{\partial T}{\partial X}\right), \tag{3.3}$$

$$P = nk_{\rm B}T. \tag{3.4}$$

In this analysis, we neglect He atoms for simplicity. The effect of chemical evolution can be included in the nonequilibrium cooling and heating function. For the range of temperatures and densities considered, the chemical timescale is longer than the cooling timescale. We use the non-equilibrium chemical fraction just after shock heating.

3.2 The Unperturbed State

To express isobarically contracting background, we use the comoving coordinate x related to the real coordinate X,

$$x = X/a(t), \tag{3.5}$$

where $a(t)$ is the contraction parameter. In these contracting coordinates, the velocity field can be written as

$$v = \dot{a}x + v_1, \tag{3.6}$$

where v_1 is the perturbed velocity relative to the contraction.

Lagrangian differentiation of physical variable f is

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \left(v\frac{\partial}{\partial X}\right) f = \left(\frac{\partial f}{\partial t}\right)_x + \frac{1}{a}\left(v_1\frac{\partial}{\partial x}\right) f. \tag{3.7}$$

In comoving coordinates, the one-dimensional hydrodynamic equations are

$$\dot{n} + \frac{\dot{a}}{a}n + \frac{1}{a}\frac{\partial}{\partial x}(nv_1) = 0, \tag{3.8}$$

$$\ddot{a}x + \dot{v_1} + \frac{\dot{a}}{a}v_1 + \frac{1}{a}\left(v_1\frac{\partial}{\partial x}\right) v_1 + \frac{1}{m_{\rm H}na}\frac{\partial P}{\partial x} = 0, \tag{3.9}$$

$$\dot{P} + \frac{1}{a}\left(v_1\frac{\partial}{\partial x}\right) P - \gamma\frac{P}{n}\left[\dot{n} + \frac{1}{a}\left(v_1\frac{\partial}{\partial x}\right) n\right] + (\gamma-1)(n\Lambda - n\Gamma) = \frac{\gamma-1}{a^2}\frac{\partial}{\partial x}\left(K\frac{\partial T}{\partial x}\right). \tag{3.10}$$

If we set the perturbed velocity v_1 equals to zero in the above equations, the equations governing isobarically contracting background are

$$\dot{n} + \frac{\dot{a}}{a}n = 0, \tag{3.11}$$

$$\ddot{a}x + \frac{1}{m_{\rm H}na}\frac{\partial P_0}{\partial x} = 0, \tag{3.12}$$

$$\gamma\frac{\dot{a}}{a} + (\gamma-1)\frac{\Lambda-\Gamma}{k_{\rm B}T} = 0. \tag{3.13}$$

Equation 3.11 leads $n \propto a^{-1}$. Temperature is proportional to a because unperturbed gas is isobaric. From equation (3.13), we obtain the approximate solution for $a(t)$ as

$$a(t) = a(0)(1 - t/\gamma\tau_{\rm net}), \tag{3.14}$$

where $\tau_{\rm net}$ is a net cooling time defined by $\tau_{\rm net}^{-1} = (\gamma-1)(\Lambda-\Gamma)/k_{\rm B}T$. This approximation is valid as long as variations of density and temperature remain small, i.e. $0 < t \ll \gamma\tau_{\rm net}$. Equation (3.12) gives $\partial P_0/\partial x = 0$. Thus, we consider uniform unperturbed state, $\partial n_0/\partial x = 0$. The isobarically cooling gas is characterized by a temperature $T_0(t)$ and a number density $n_0(t)$ which are functions of time, while the pressure P_0 is constant.

3.3 The Linearized Equations

We now examine the effect of small perturbations:

$$n(x,t) = n_0(t)(1+n_1(t)\exp(i\kappa x)), \tag{3.15}$$
$$P(x,t) = P_0(1+P_1(t)\exp(i\kappa x)), \tag{3.16}$$
$$T(x,t) = T_0(t)(1+T_1(t)\exp(i\kappa x)), \tag{3.17}$$
$$v_1(x,t) = v_1(t)\exp(i\kappa x). \tag{3.18}$$

The first-order equations for T_1, P_1, n_1, and v_1 are

$$\dot{n_1} + \frac{i\kappa}{a}v_1 = 0, \tag{3.19}$$
$$\dot{v_1} + \frac{\dot{a}}{a}v_1 + c_{\rm s}^2\frac{i\kappa}{a}P_1 = 0, \tag{3.20}$$
$$\dot{P_1} + \frac{\dot{a}}{a}P_1 - \gamma\dot{n_1} + \frac{n_1}{\tau_{\rm net}} + \frac{s_{\rm T}T_1 + s_\rho n_1}{\tau_{\rm cool}} - \frac{r_{\rm T}T_1 + r_\rho n_1}{\tau_{\rm heat}} = -\ell c_{\rm s}\frac{\kappa^2}{a^2}T_1, \tag{3.21}$$
$$P_1 = n_1 + T_1, \tag{3.22}$$

where

$$s_{\rm T} = \partial\ln\Lambda/\partial\ln T, \tag{3.23}$$
$$s_\rho = \partial\ln\Lambda/\partial\ln n, \tag{3.24}$$
$$r_{\rm T} = \partial\ln\Gamma/\partial\ln T, \tag{3.25}$$
$$r_\rho = \partial\ln\Gamma/\partial\ln n, \tag{3.26}$$
$$\tau_{\rm cool}^{-1} = (\gamma-1)\Lambda/k_{\rm B}T, \tag{3.27}$$
$$\tau_{\rm heat}^{-1} = (\gamma-1)\Gamma/k_{\rm B}T. \tag{3.28}$$

We adopt the collision mean free path, ℓ, instead of thermal conduction coefficient $K = nk_{\rm B}\ell c_{\rm s}/(\gamma-1)$. The term $\dot{a}/a$ remains constant $(-1/\gamma\tau_{\rm net})$ as long as $t \ll \gamma\tau_{\rm net}$. We assume that the perturbation is proportional to $\exp(\sigma t)$ and obtain the dispersion relation as

$$\left[\sigma^2 - \frac{2\sigma}{\gamma\tau_{\rm net}} + (kc_{\rm s})^2\right]\left(\sigma + \frac{s_{\rm T}-1}{\tau_{\rm cool}} - \frac{r_{\rm T}-1}{\tau_{\rm heat}} + \ell c_{\rm s}k^2\right)$$
$$-(kc_{\rm s})^2\left[(1-\gamma)\sigma + \frac{s_\rho}{\tau_{\rm cool}} - \frac{r_\rho}{\tau_{\rm heat}}\right] = 0, \tag{3.29}$$

where $k = \kappa/a$. This cubic equation for σ has a real root and two complex conjugate roots. The former root is called condensation mode, and the latter roots are called wave mode. Following Field (1965), we call the most unstable mode the condensation mode. In Figure 3.1, we show an example of dispersion relation of the condensation mode.

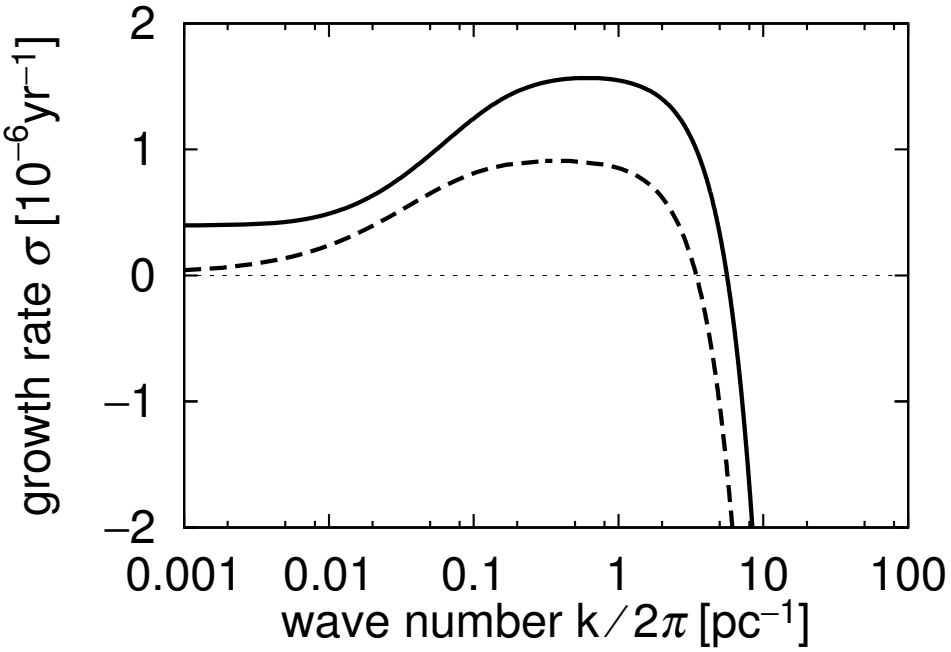


Figure 3.1: Dispersion relation for the condensation mode of the isobarically contracting uniform gas with $n = 10^{0.25}$, $T = 10^{3.0}$ (*solid line*). The case of thermal equilibrium unperturbed state is also shown (*dashed line*).

4

EVOLUTION OF INTERSTELLAR SHOCK-COMPRESSED LAYERS

In this chapter, we study the basic processes in the evolution of the ISM with a strong shock wave by means of one-dimensional hydrodynamic calculations with nonequilibrium thermal and chemical processes. The processes we include in our calculations are photoelectric heating from small grains and polycyclic aromatic hydrocarbons (PAHs), heating and ionization by cosmic rays and X-rays, heating by H_2 formation and destruction, atomic line cooling from hydrogen Lyα, C II, O I, Fe II, and Si II, rovibrational line cooling from H_2 and CO, and atomic and molecular collisions with grains. Effects of self-gravity and the magnetic field are not treated in this chapter. In the next section, we explain the details of our numerical methods. Results are presented in §4.2. In §4.3, we discuss the formation of small molecular cloudlets.

4.1 NUMERICAL METHODS

4.1.1 Hydrodynamic Equations

To study the thermal and dynamical evolution of a shock-compressed layer, we use a one-dimensional plane parallel numerical method. The hydrodynamics module of our scheme is based on the second-order Godunov method. We use the result of Riemann problem at each cell interface to calculate numerical fluxes, without introducing any artificial viscosity, which characterizes the Godunov method. A higherorder Godunov method has been the state-of-the-art numerical method in last two decades (see e.g., van Leer 1997) and widely used in an astrophysical context (see e.g., Truelove et al. 1998). The hydrodynamic equations for the system are written as follows:

$$\frac{d\rho}{dt} + \rho\frac{\partial v}{\partial X} = 0, \tag{4.1}$$

$$\rho\frac{dv}{dt} = -\frac{\partial P}{\partial X}, \tag{4.2}$$

$$P = \left(1.1 + x_{\rm e} - \frac{x_2}{2}\right) n k_{\rm B} T, \tag{4.3}$$

$$\rho = 1.4 m_{\rm H} n, \tag{4.4}$$

$$\frac{1}{\gamma - 1}\frac{dP}{dt} - \frac{\gamma}{\gamma - 1}\frac{P}{\rho}\frac{d\rho}{dt} = n\Gamma - n\Lambda + \frac{\partial}{\partial X}K\frac{\partial T}{\partial X}. \tag{4.5}$$

In the above equations, n is the number density of hydrogen nuclei, ρ is the mass density of the gas, v is the velocity of fluid elements, P is the gas pressure, and K is the coefficient of thermal conductivity. We have assumed an abundance of He atom is $0.1n$. For the range of temperatures and ionized fractions considered, the dominant contribution to thermal conductivity is that due to neutral atoms, for which $K = 2.5 \times 10^3 T^{1/2}\,{\rm cm^{-1}K^{-1}s^{-1}}$ (Parker 1953). We use the ratio of the specific heats $\gamma = 5/3$ for simplicity. The value of Γ and Λ are the heating and cooling rate per hydrogen nucleus, respectively. These rates depend on number density n, temperature T, electron fraction $x_{\rm e} = n({\rm e^-})/n$, H_2 fraction $x_2 = 2n({\rm H_2})/n$, and CO fraction $x_{\rm CO} = n({\rm CO})/n$. The gas behind the shock front is in a chemical nonequilibrium state. Therefore we solve three chemical equations for $x_{\rm e}$, x_2, and $x_{\rm CO}$ and hydrodynamic equations simultaneously.

4.1.2 Heating and Cooling Processes

Wolfire et al. (1995, hereafter W95) calculated the thermal equilibrium state in the neutral atomic phase. We follow their calculation for $n < 10^3\,{\rm cm^{-3}}$. We also calculate molecular processes for $10^3\,{\rm cm^{-3}} < n < 10^6\,{\rm cm^{-3}}$ (see Figure 4.1). Contributions to the heating of an interstellar cloud are from the photoelectric emission from small grains and PAHs, ionization by cosmic rays and soft X-rays, and the formation and photodissociation of H_2. Following W95, the local far-ultraviolet (FUV) field is set to 1.7 times Habing's estimate (i.e., $G_0 = 1.7$). The cooling function is dominated by line emission from H, C, O, Si, and Fe, by rovibrational lines from H_2 and CO, and by atomic and molecular collisions with dust grains. To solve these thermal processes in nonequilibrium, we solve a set of three time-dependent equations for the ionization and recombination of hydrogen and for the formation and dissociation of H_2 and CO. Self-shielding effects are also included in the calculation of H_2 photodissociation. The details of these processes are presented in Chapter 2. Figure 4.1 shows the equilibrium temperature, pressure, and chemical fractions as functions of number density. Important physical timescales are also shown in Figure 4.1d.

4.1.3 Initial and Boundary Conditions

The strong shock wave is characterized by Mach number. To analyze the evolution of ISM swept up by a strong shock wave, we set up a plane-parallel cloud which collides against a rigid wall. Figure 4.2 shows the schematic configuration of our calculation. This configuration also corresponds to a face-on collision between two identical clouds.

We have generated a grid of calculations for varying initial densities and collision velocities. The typical models are summarized in table 4.1. Initial temperatures and chemical fractions are taken to be the value of the thermal and chemical equilibrium state (see Fig.

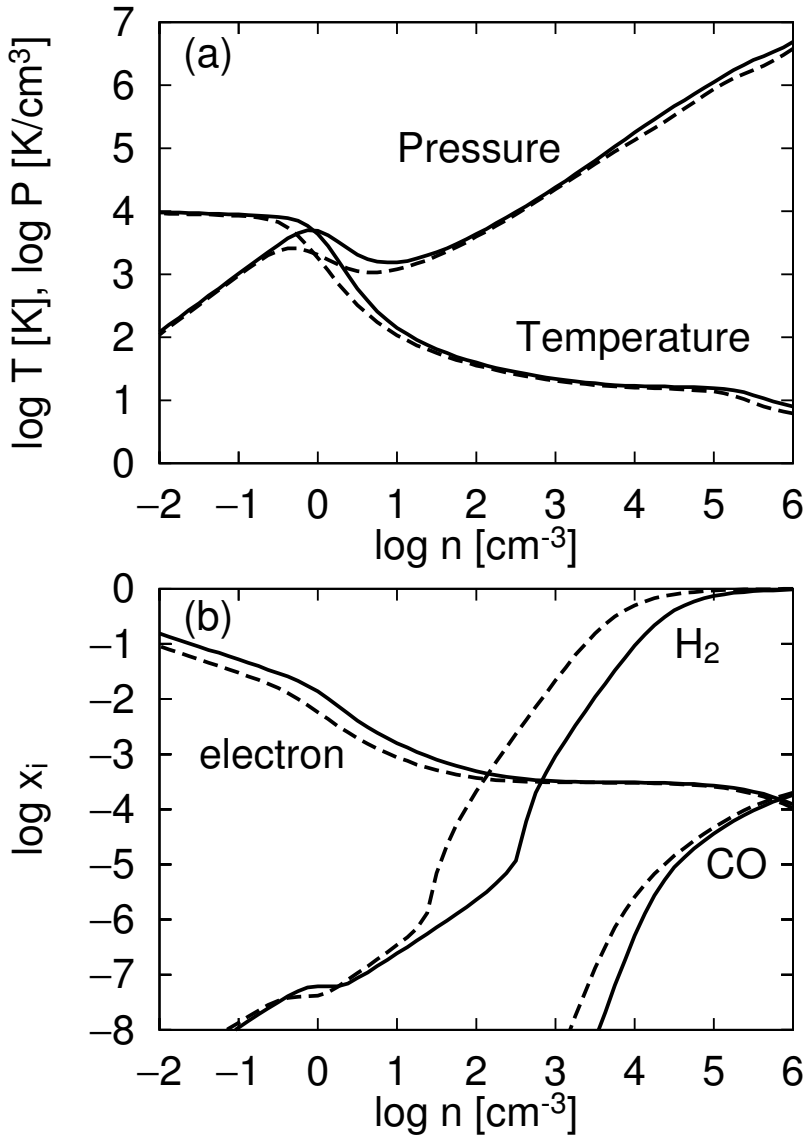
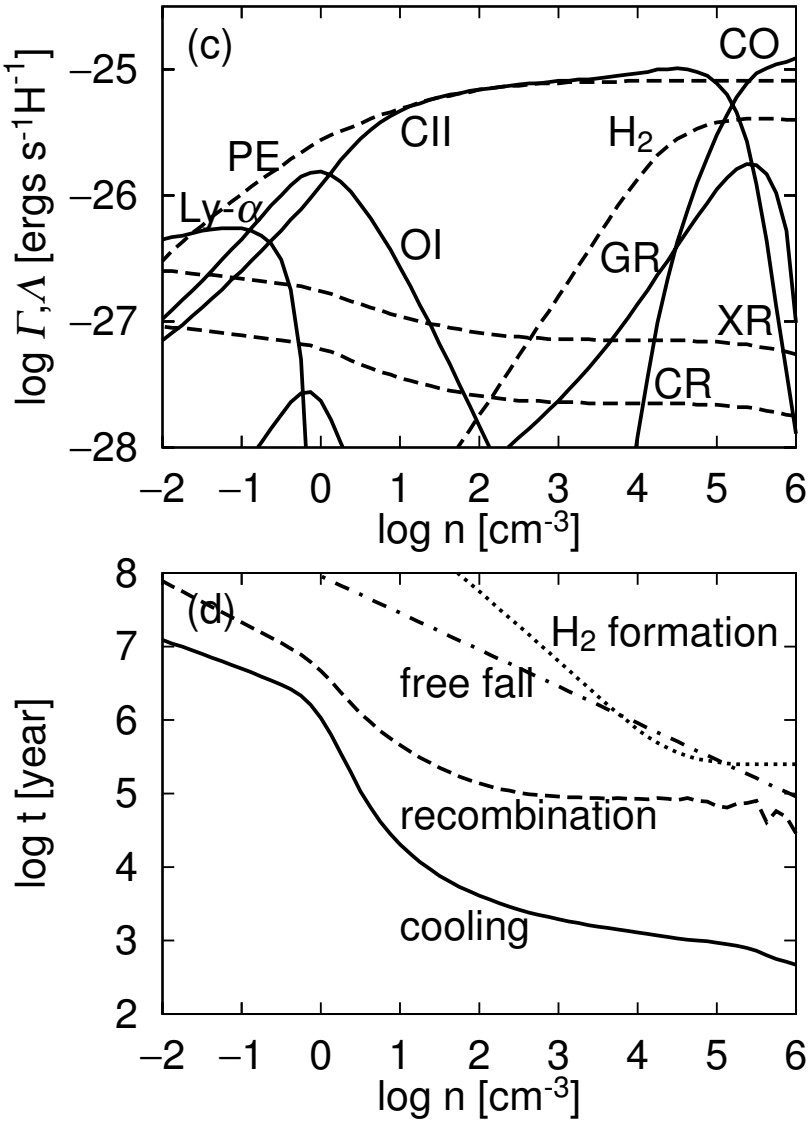


Figure 4.1: (a) Equilibrium temperature and pressure. An absorbing column is $10^{19}\,\mathrm{cm}^{-2}$ (*solid lines*), $10^{20}\,\mathrm{cm}^{-2}$ (*dashed lines*). (b) Chemical fractions as functions of number density. (c) Heating (*dashed lines*) and cooling (*solid lines*) rates per hydrogen nucleus in equilibrium condition corresponding to panel a (*solid lines*). Heating processes are, photoelectric heating from small grains and PAHs (PE), X-ray (XR), Cosmic-ray (CR), and H_2 formation/destruction heating (H_2). Cooling processes are CII fine-structure (CII), OI fine-structure (OI), Hydrogen Lyman-α (Ly-α), CO rotation/vibration line (CO), and Atomic and molecular collisions with dust grains (GR). (d) Comparison of various timescales, cooling time (*solid line*), recombination time (*dashed line*), free fall time (*dot-dashed line*), and H_2 formation time (*dotted line*) which is defined by $(Rn_{\rm HI})^{-1}$.


(c)
CO
CII
H2
PE
Ly-α
OI
GR
XR
CR
log Γ,Λ [ergs s-1H-1]
log n [cm-3]
(d)
H2 formation
free fall
recombination
cooling
log t [year]
log n [cm-3]

4.1). The velocity difference, $V_{\rm d}$, defined by upstream velocity minus downstream velocity, corresponds to the range of Mach number $M = 2 - 12$. This range of Mach number in the WNM ($T \sim 8000$ K) corresponds to the sweeping-up speed of momentum-driven SNRs. External FUV and X-ray radiation penetrate from both sides of two identical clouds. The extinction of the radiation is evaluated by the cumulative column density of each grid. The largest total column density we considered is that of the standard H I cloud (10^{19} – 10^{20} cm^{-2}).

4.2 RESULTS

4.2.1 Shock Propagation into WNM

In this section, we analyze the model W6 where a shock wave propagates into WNM. Figure 4.3 shows the result of the calculation. Snapshots at $t = 8.0 \times 10^4$, 1.3×10^5, 2.5×10^5, and 2.9×10^6 yr are shown. The horizontal axis in logarithmic scale denotes the distance from the rigid wall. Figure 4.3a shows the evolution of pressure. The shock-compressed layer is almost isobaric. The shock front propagates from left to right. Figure 4.3b shows the evolution of temperature. Lyα cooling is so efficient that the temperature of post-shock gas quickly decreases to the pre-shock value. In §4.2.2, we analyze the shock front in detail. Behind the shock front, cooling dominates heating because density is larger and temperature decreases monotonically. Around $t \simeq 10^5$ yr, temperature starts to decline rapidly because of thermal instability. This instability is mainly driven by O I (63 μm) and C II (157 μm) line cooling. Eventually temperature attains a thermal equilibrium value $T \simeq 19$ K at $n \simeq 2.5 \times 10^3\,{\rm cm}^{-3}$. The main coolant of collapsed the layer is C II (see Fig. 4.1c). Figure 4.3c shows the evolution of number density. A thin layer is formed by thermal instability in shock-compressed layer. The width of the thermally collapsed layer is much shorter than the shock-compressed layer. However, the column density of the thermally collapsed layer becomes comparable after $t = 2.6 \times 10^5$ yr (Fig. 4.3c, *dotted line*). Figure 4.3d shows the evolution of velocity. Figure 4.3e shows the evolution of electron fraction. High shock temperature raises the ionization degree in the postshock gas. The timescale of recombination of electrons in postshock layer is larger than the timescale of gas cooling. Figure 4.3f shows the evolution of the H_2 number fraction. Because the timescale of H_2 formation is much larger than cooling timescale (see Fig. 4.1d), this thermal instability is not driven by H_2 cooling. In the highest density region at the final time step (*dot-dashed line*), 1.3% of the hydrogen is in H_2 and 0.0002% of the carbon is in CO. Note that the preshock H_2 number fraction at the final time step (*dot-dashed line*) is larger than the initial fraction because H_2 photodissociation radiation from the left hand side is shielded by large H_2 column density in the collapsed layer in this one-dimensional plane-parallel model.

Table 4.1. MODELS.

Model	$n_{\rm i}$[a]	$T_{\rm i}$[b]	$x_{\rm ei}$[c]	$N_{\rm H}$[d]	M[e]	$V_{\rm d}$[f]	$t_{\rm f}$[g]	$n_{\rm f}$[h]	$T_{\rm f}$[i]	$P_{\rm f}/k_{\rm B}$[j]	$x_{\rm 2f}$[k]
W2	0.1	8491	0.03	1(20)[l]	2	18	7.0(6)	1.73	6353	3.3(3)	9.5(-8)
W6 [m]	0.1	8491	0.03	1(20)	6	54	2.9(6)	2.5(3)	18.7	5.2(4)	0.013
W6L	0.1	8912	0.05	1(19)	6	54	2.8(5)	2.5(3)	19.0	5.2(4)	5.6(-5)
W12	0.1	8491	0.03	1(20)	12	109	1.45(6)	1.2(4)	16.6	2.0(5)	0.21
I12	0.1	8491	1.00	1(20)	12	109	1.45(6)	1.2(4)	16.6	2.0(5)	0.21
C2	10	107	8.7(-4)	1(20)	2	2.1	7.0(5)	2.8(2)	28.5	8.8(3)	5.1(-5)
C10 [n]	10	107	8.7(-4)	1(20)	10	10	1.0(5)	1.4(4)	16.7	2.5(5)	0.049

[a]The initial number density of hydrogen nuclei in $\rm cm^{-3}$.

[b]The initial temperature in K.

[c]The initial electron fraction.

[d]The total column density of hydrogen nuclei in $\rm cm^{-2}$.

[e]The Mach number in the center of mass frame.

[f]The velocity difference in km $\rm s^{-1}$, defined by upstream velocity minus downstream velocity which we set to be zero.

[g]The final time in years.

[h]The maximum density in $\rm cm^{-3}$ at final time step.

[i]The temperature in K in the maximum density regions at the final time step.

[j]The pressure in K $\rm cm^{-3}$ in the maximum density regions at final time step.

[k]The H_2 fraction in the maximum density region at the final time step.

[l]The notation a(b) means $a \times 10^b$

[m]Details of this model are discussed in §4.2.1

[n]Details of this model are discussed in §4.2.4

4.2.2 Structure across the Shock Front

In this subsection, we analyze the detailed structure across the shock front which propagates into the WNM. The same calculations as model W6 at $t = 1 \times 10^4$ are shown in Figure 4.4. Dashed lines show the calculation which used the same initial grid spacing as model W6. We also calculated the model with 320 times higher spatial resolution (*solid lines*) to resolve detailed structure of the shock front. Dotted lines denote the same calculation without cooling and heating processes. The width of the shock front is on the order of the collision mean free path, $\ell = 1/n\sigma_{col} \approx 0.0003/n$ pc. For a strong shock without cooling, postshock temperature is determined by the Rankine-Hugoniot relation as $T_2 = (\gamma - 1)mV_{\rm d}^2/2k_{\rm B}$. On the other hand, shock temperature with cooling is estimated by the following equation:

$$\frac{k_{\rm B}T_{\rm s}}{\gamma - 1} + E_{\rm loss} = \frac{1}{2}mV_{\rm d}^2, \tag{4.6}$$

where $E_{\rm loss} = \int n(\Lambda - \Gamma)\, dt \approx n\Lambda_{{\rm Ly}\alpha}(T_{\rm s})\ell/V_{\rm d}$. This analytic estimation provides $\log T_{\rm s}$ [K] = 4.8 in this model. Even the lowest resolution model can provide the same values of physical quantities in the postshock region.

Radiative precursor which is not treated in our models, can change preshock temperature and ionization. Shull & McKee (1979) have calculated the ionization of the preshock gas which is caused by the diffuse radiation from the postshock gas in their models of interstellar shock waves with v_s = 40 – 130 km/s. They have concluded that sufficiently fast shocks ($v_{\rm s} > 110$ km/s) completely ionize the preshock gas. To study this effect, we also calculated the model with fully ionized preshock gas (model I12) and found no significant difference between the model I12 and W12. Therefore, we neglect the effect of radiative precursor in the rest of this thesis.

4.2.3 Thermal Instability of Isobarically Cooling WNM

Figure 4.5 shows the evolution of the fluid element which has the maximum density at each time step. Density, temperature and pressure evolve on the dashed lines from left to right. Solid and thick lines correspond to the thermally stable equilibrium. Dotted lines correspond to the thermally unstable equilibrium. The preshock gas is a thermally stable WNM ($n = 0.1\,{\rm cm}^{-3}$, $T = 8000$ K). In Chapter 3, we investigate the stability of isobarically contracting gas by linear perturbation theory to analyze the evolution on the dashed lines. We have used the postshock values of hydrodynamic calculation as the chemical compositions in the linear analysis ($x_{\rm e} = 0.1$, $x_2 = 10^{-6.3}$, $x_{\rm CO} = 10^{-23}$). Because the chemical reaction timescale is longer than the cooling timescale. The shaded area in Figure 4.5b denotes thermally unstable region determined by the linear analysis. The larger electron fraction makes cooling rate larger, so that the unstable region is wider than the unstable region of equilibrium gas. This shaded region shows that thermal instability is inevitable when WNM cools into CNM.

Next, we discuss about the characteristic length and timescale of the thermal instability. The linear growth rate of thermal instability as a function of perturbation wavelength is calculated with a given temperature and density. The growth time of the instability is shown in Figure 4.6. In this figure, the vertical axis denotes the unperturbed temperature. The value of the constant pressure, $P_c/k_B = 5 \times 10^4\,\mathrm{K/cm^3}$, is adopted from the result of our non-linear calculation. The density can be deduced from the relation, $n = P_c/k_B T$. The horizontal axis denotes a quarter of the perturbation wavelength. Contours of growth time are depicted in this wavelength-temperature plane. In Figure 4.6b, the temperature evolution of our non-linear calculation (Fig. 4.3b) is superposed upon the growth rate contours (Fig. 4.6a). Thermal instability occurs at temperature less than 7300 K. The instability becomes drastic at temperature below 1000 K, which corresponds to the growth time less than 10^4 yr. In this way we can understand the collapse of the layer in terms of thermal instability. The shortest wavelength of unstable perturbation is $\lambda_{\min} \sim 8 \times 10^{-5}\mathrm{pc} \approx 16$ AU. The thickness of the thermally-collapsed layer increases with time through the accretion of gas.

4.2.4 Shock Propagation into CNM

In this section, we analyze the model C10, where shock wave propagates into CNM. Although Smith (1980) has done a similar calculation, his calculation does not have sufficient dynamic range to resolve the thermally collapsing layer which is of our interest. Figure 4.7 shows the results of our calculation where the Mach number $M = 10$, which corresponds to the velocity difference of 10 km/s. Snapshots at $t = 3.1 \times 10^3$, 5.2×10^4, 5.7×10^4, and 8.8×10^4 yr are shown. The shock front is almost adiabatic. The postshock pressure can be estimated by the Rankine-Hugoniot relation as follows:

$$\frac{P_2}{P_1} = 1 + \frac{\gamma(\gamma+1)}{4}M^2 + \frac{\gamma}{4}M\sqrt{(\gamma+1)^2M^2+16}, \tag{4.7}$$

where M is the Mach number in the center of mass frame. When M equals to 10, the postshock pressure is 224 times higher than the ambient pressure. Details of postshock structure are more complicated than those in the WNM case. The postshock gas, of which temperature and density are determined by the adiabatic shock condition, is thermally unstable (see Fig. 4.8b). The shock front decelerates after postshock layer collapses. After the thermally collapsed layer is formed, accretion shock is formed inside the shock-compressed layer. Accretion shock propagates into compressed layer. In the highest density region at the final time step, 4.9% of the hydrogen is in H_2 and 0.02% of the carbon is in CO. Note that the abundances of these molecules are still increasing because the H_2 formation timescale is much longer than the collapse timescale $t \simeq 10^5$ yr (see Fig. 4.1d).

4.2.5 Thermal Instability of Isobarically Cooling CNM

Figure 4.8 shows the evolution of the fluid element which has the maximum density at each time step. The shaded area of Figure 4.8b denotes thermally unstable region. In this case of

linear analysis (Chapter 3), chemical compositions in thermal and chemical equilibrium states of each density are used. This Figure shows how thermally stable CNM becomes unstable. The growth time of the instability is shown in Figure 4.9. In Figure 4.9b, we superpose the temperature evolution of our non-linear calculation (Fig. 4.7b) upon the growth rate contours (Fig. 4.9a). The shortest wavelength of unstable perturbation is $\lambda_{\rm min} \sim 2.5 \times 10^{-5}$pc ≈ 5 AU.

4.2.6 Results from Various Initial Parameters

Calculations are done at nine different initial densities ($10^{-2} < n < 10^2$), and 11 velocity differences ($2 <$ Mach number < 12). The column density is 10^{20} cm^{-2}. We follow the calculation until $t = 10^5$ – 10^6 yr. The number fraction of H_2 in the highest density region is shown in Figure 4.10. The horizontal axis denotes the initial number density of hydrogen nuclei. The vertical axis denotes the velocity difference. Solid lines denote the 8 % boundary of the H_2 number fraction. Higher velocity produces more abundance of H_2 in the thermally collapsed layer. Note that the fragmentation of the thermally collapsed layer will change the column density of the fragments, which change the efficiency of H_2 self-shielding. Therefore, the amount of H_2 in the fragments would be different from the value obtained in this paper. To determine the realistic abundance of H_2, two or three-dimensional hydrodynamic calculation is required.

4.3 DISCUSSION

Our hydrodynamic calculations show that thermal instability produces the collapsed layer with a considerable amount of H_2. In this section, we discuss the consequence of this instability.

4.3.1 Fragmentation of the Thermally Collapsed Layer

The shock propagation into ISM can produce the thermally collapsing gas inside the shock-compressed layer. The initial thickness of the thermally collapsed layer is dozens of AU and agrees with the prediction by the linear analysis. The thickness of the thermally collapsed layer increases with time through the accretion of gas. The thermally collapsing layer is dynamically unstable also in the y- and z-direction, so that perturbations in the layer will grow and lead to fragmentation of the layer. We expect that this layer will break up into very small cloudlets which have different translational velocities. This velocity dispersion of cloudlets produced by the passage of supernova shocks may make an origin for the observed interstellar "turbulent" velocities. Extremely high pressure in the tiny-scale structure observed in H I clouds (Heiles 1997) is consistent with our shock propagation model.

4.3.2 Molecular Clouds in The Galaxy

It seems difficult to detect these cloudlets in emission lines of CO molecules (see e.g., Liszt & Wilson 1993, Liszt 1994). Many shell-like or filamentary structures in the H I 21-cm observation maps (Hartmann & Burton 1997) are reminiscent of overlapping shells of old SNRs. If these structures in H I maps are really the results of SNR shocks, many tiny molecular cloudlets should be hidden in the shell and should be ubiquitous in the Galaxy. McKee & Ostriker (1977) showed that the recycling SNRs in the Galaxy would overlap before they are dissipated. The overlapping filamentary region may produce larger clouds.

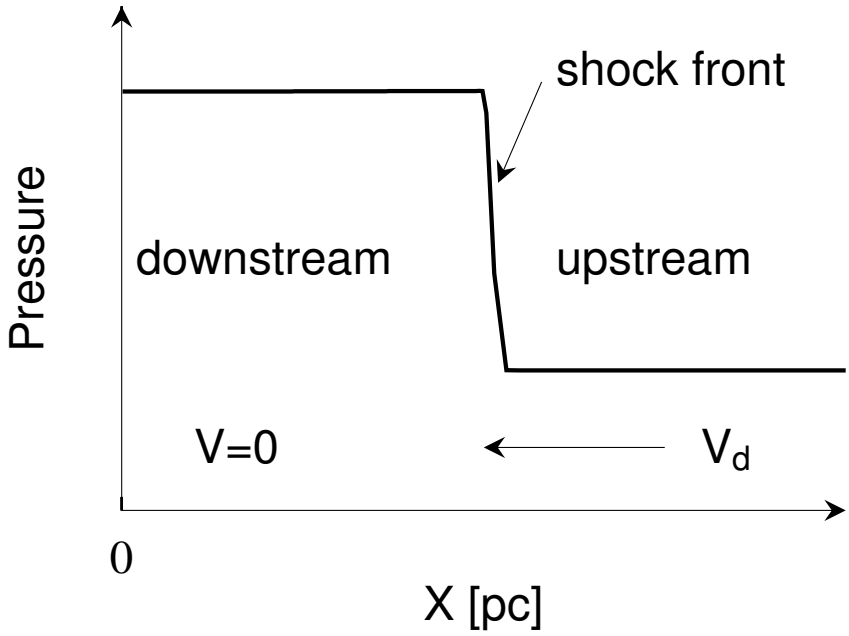


Figure 4.2: Schematic configuration of our calculation. The shock front propagates from left to right. The velocity difference, $V_{\rm d}$, is defined by upstream velocity minus downstream velocity which we set to be zero.

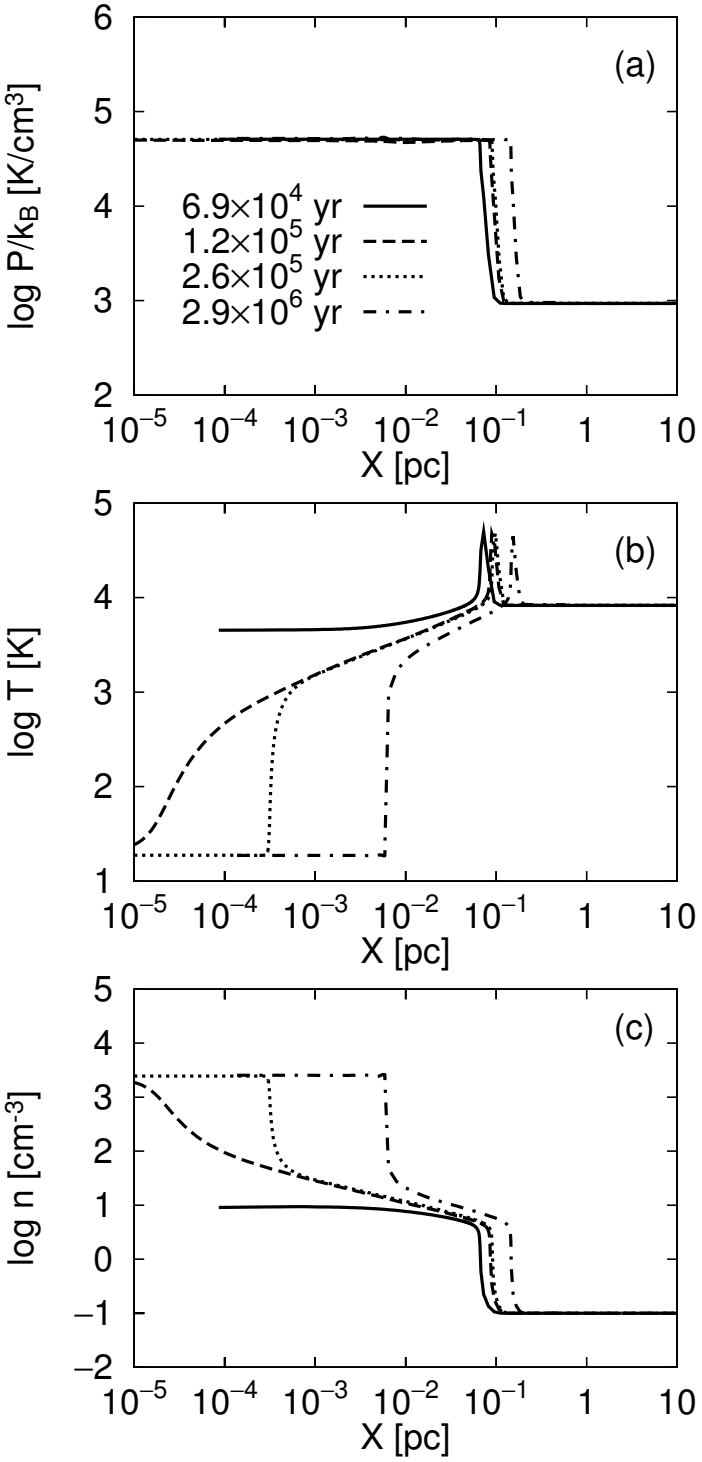
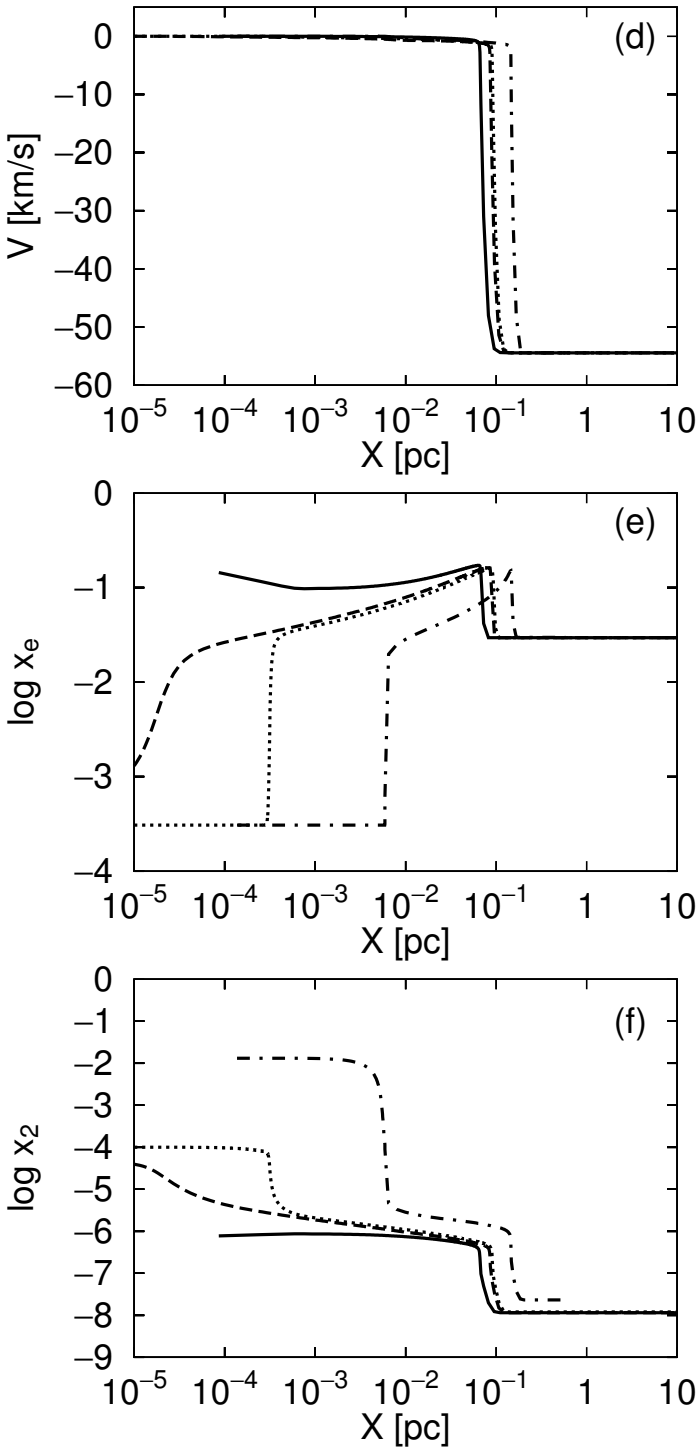


Figure 4.3: Evolution of the shock propagation into the WNM. The figures show (a) pressure, (b) temperature, (c) number density of hydrogen nuclei, (d) velocity, (e) electron number fraction, and (f) H_2 number fraction. Solid lines denote the compressed layer at $t = 8.0 \times 10^4$ yr, dashed lines at $t = 1.3 \times 10^5$ yr, dotted lines at $t = 2.5 \times 10^5$ yr, and dot-dashed lines at $t = 2.9 \times 10^6$ yr.


(d)
V [km/s]
0
−10
−20
−30
−40
−50
−60
X [pc]
(e)
log x_e
−1
−2
−3
−4
(f)
log x_2
−5
−6
−7
−8
−9
10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 1 10

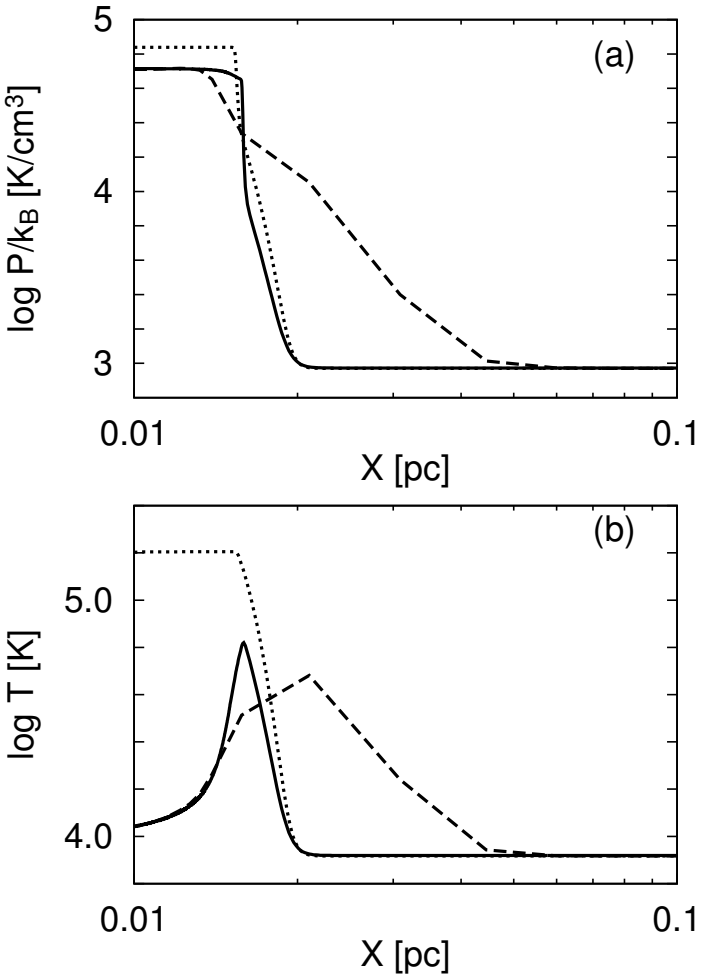
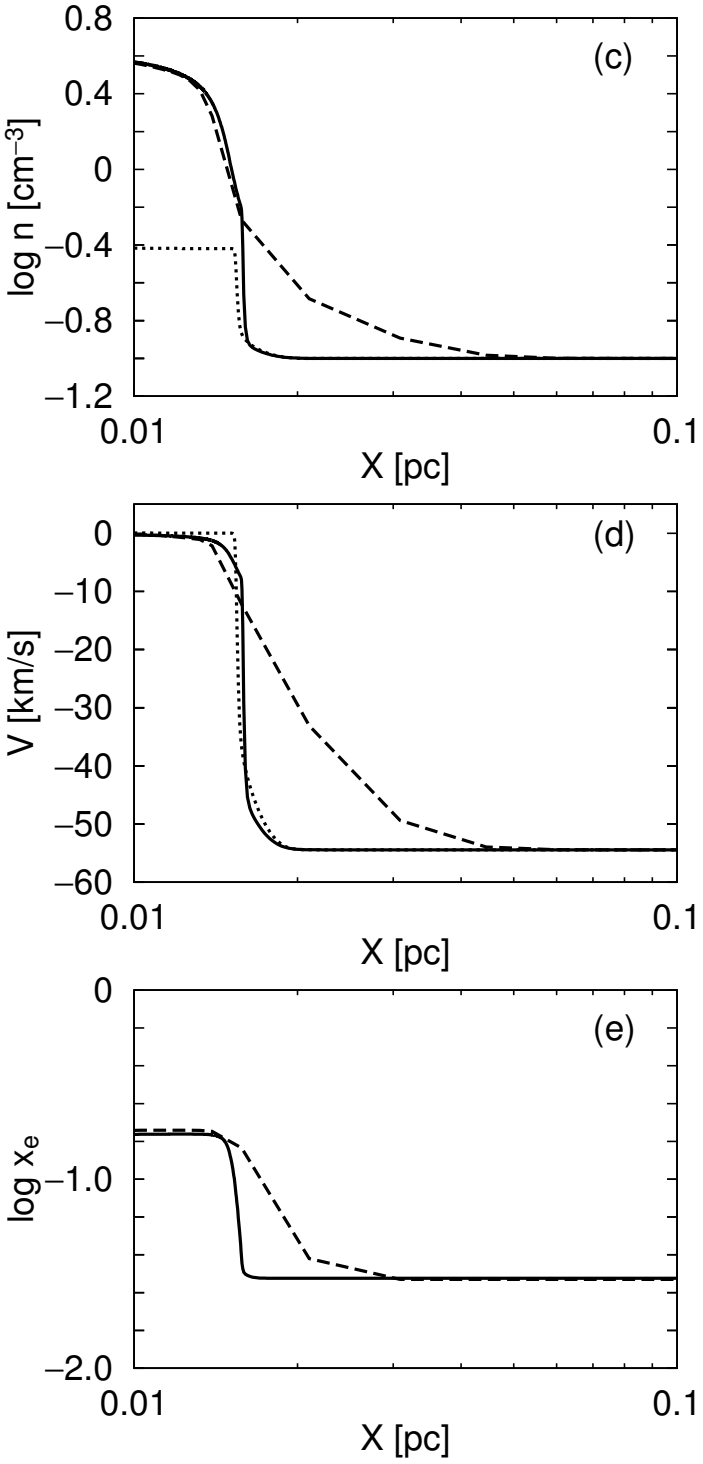


Figure 4.4: Structure across the shock front. The model W6 at $t = 1 \times 10^4$ yr is shown. The figures show (a) pressure, (b) temperature, (c) number density of hydrogen nuclei, (d) velocity, and (e) electron number fraction. Dashed lines correspond to the same spatial resolution as Fig. 4.3. Solid lines denote the calculation with 320 times higher spatial resolution than dashed lines. Dotted lines denote the calculation with 640 times higher resolution without cooling and heating.


(c)
log n [cm^{-3}]
0.8
0.4
0
−0.4
−0.8
−1.2
0.01
0.1
X [pc]
(d)
V [km/s]
0
−10
−20
−30
−40
−50
−60
0.01
0.1
X [pc]
(e)
log x_e
0
−1.0
−2.0
0.01
0.1
X [pc]

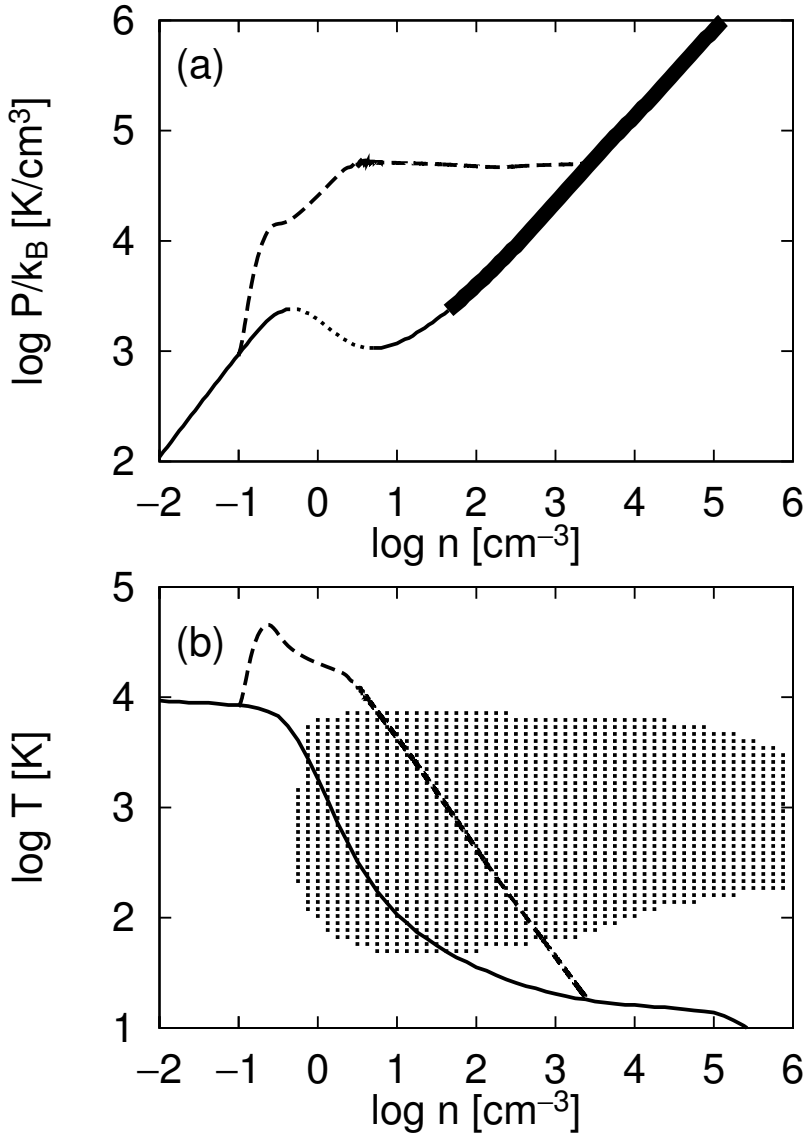


Figure 4.5: The evolution track of WNM compression on the n – P and n – T planes. Density, temperature and pressure evolve on the dashed lines from left to right. Solid and thick lines correspond to the thermally stable equilibrium. Dotted lines correspond to the thermally unstable equilibrium. In Chapter 3, we investigate the stability of isobarically contracting gas by linear perturbation theory to analyze the evolution on the dashed lines. (b) Shaded area denotes thermally unstable region determined by the linear analysis. A larger electron fraction makes cooling rate larger, so that the unstable region is wider than the unstable region of equilibrium gas. This shaded region predicts that thermal instability is inevitable when WNM cools into CNM.

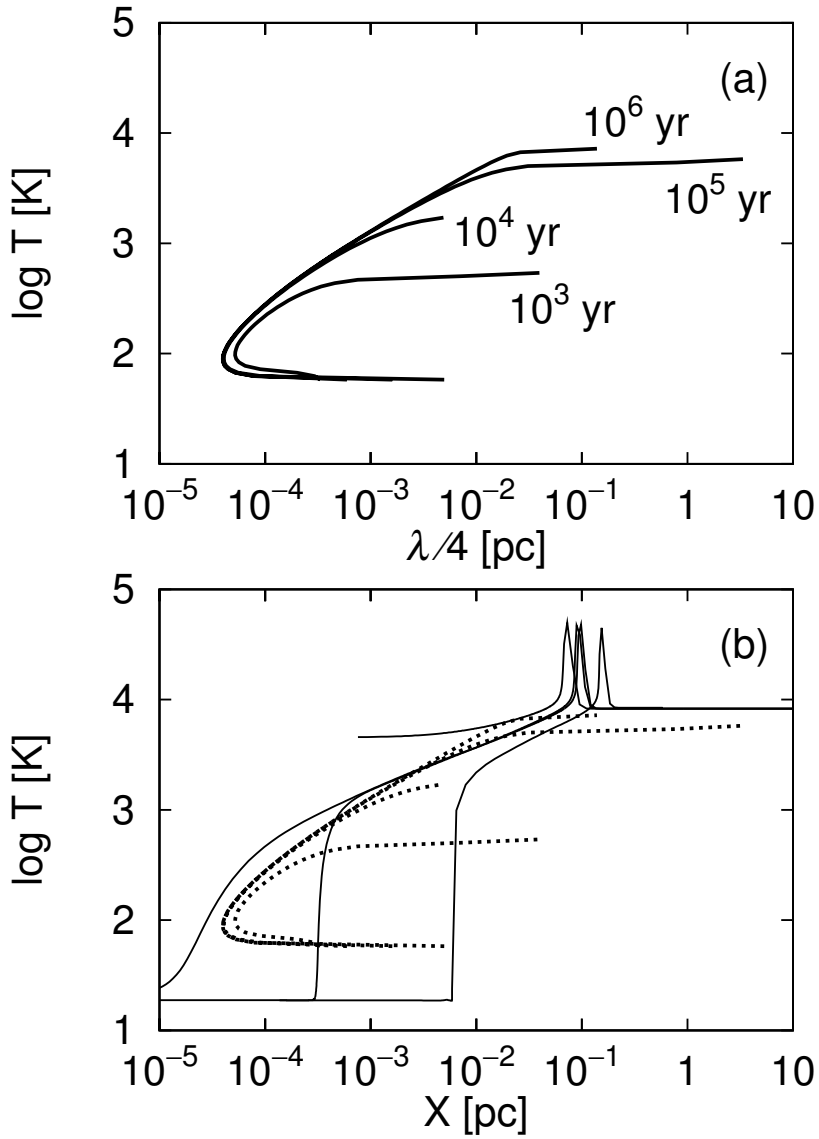


Figure 4.6: Growth time of the instability. The vertical axis denotes unperturbed temperature. The value of the constant pressure, $P_c/k_B = 5 \times 10^4\,\mathrm{K/cm^3}$, is adopted from the result of our non-linear calculation. The density can be deduced from the relation, $n = P_c/k_B T$. The horizontal axis denotes a quarter of perturbation wavelength. Contours of growth time are depicted in this wavelength-temperature plane. The shortest wavelength of unstable perturbation is $\lambda_{\rm min} \sim 8 \times 10^{-5}\,\mathrm{pc} \approx 16\mathrm{AU}$. (b) We superpose the temperature evolution of our nonlinear calculation (Fig. 4.3b) upon the growth rate contours shown in panel a.

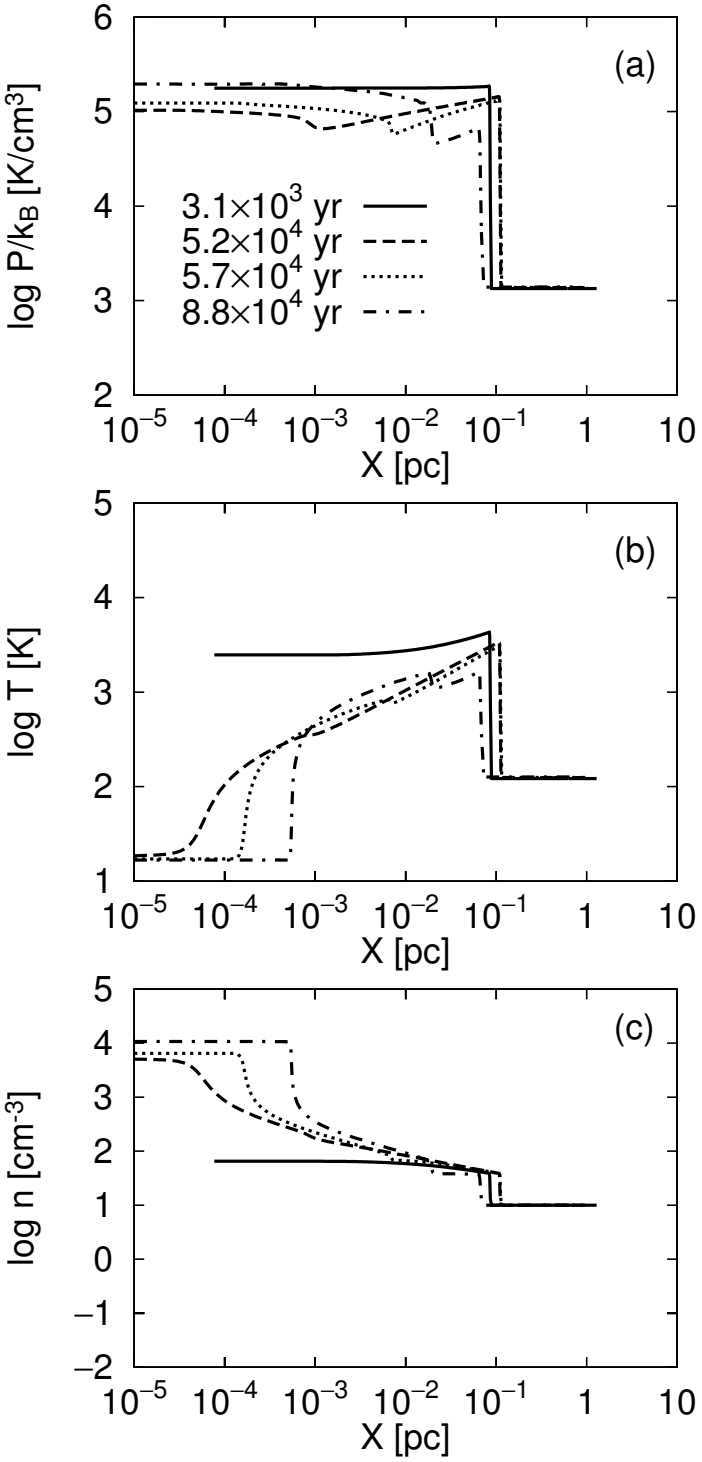
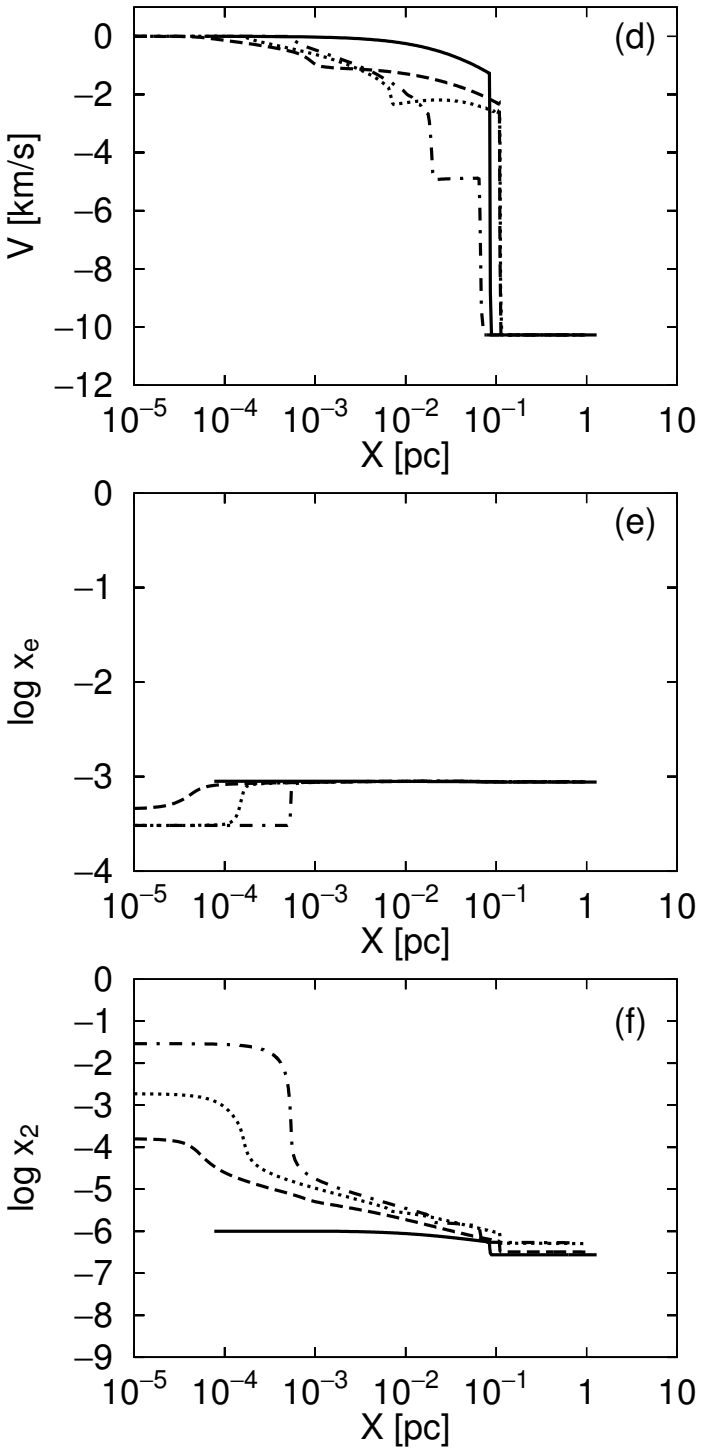


Figure 4.7: Evolution of the shock propagation into the CNM. The figures show (a) pressure, (b) temperature, (c) number density of hydrogen nuclei, (d) velocity, (e) electron number fraction, and (f) H_2 number fraction. Solid lines denote the compressed layer at $t = 3.1 \times 10^3$ yr, dashed lines at $t = 5.2 \times 10^4$ yr, dotted lines at $t = 5.7 \times 10^4$ yr, and dot-dashed lines at $t = 8.8 \times 10^4$ yr.


(d)
V [km/s]
0
−2
−4
−6
−8
−10
−12
10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 1 10
X [pc]
(e)
log x_e
0
−1
−2
−3
−4
10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 1 10
X [pc]
(f)
log x_2
0
−1
−2
−3
−4
−5
−6
−7
−8
−9
10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 1 10
X [pc]

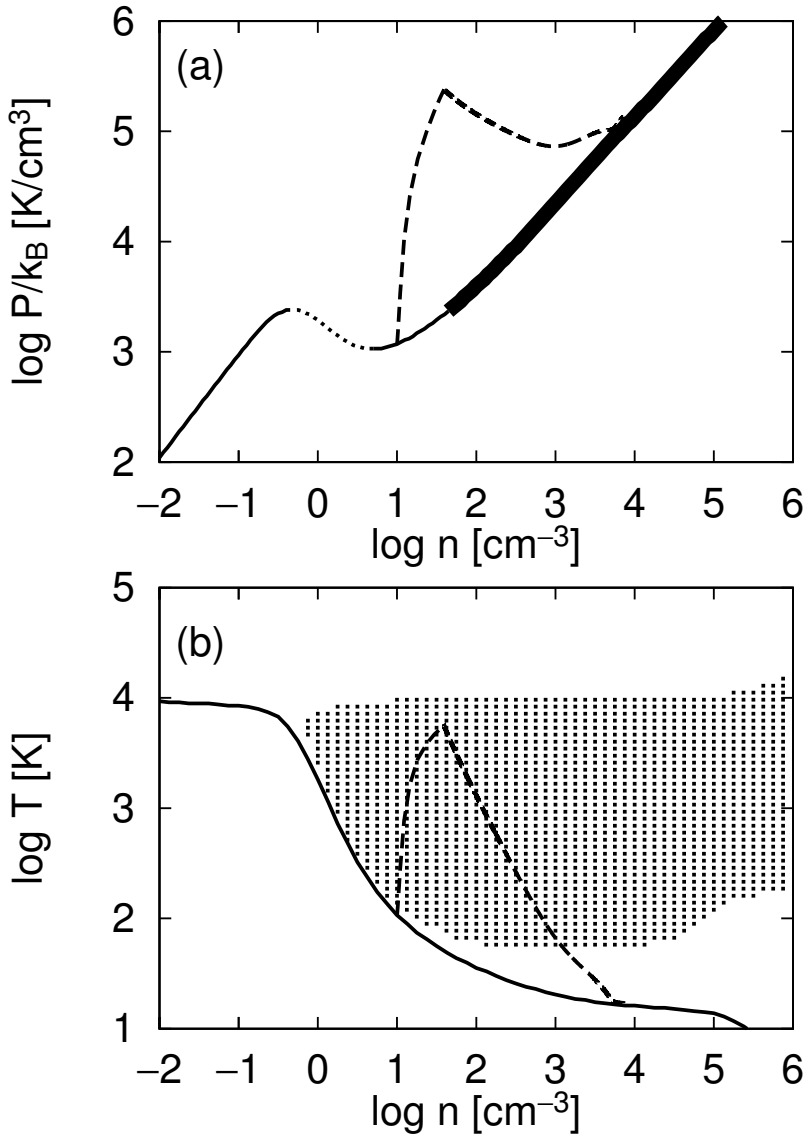


Figure 4.8: Evolution track of CNM compression on the n – P and n – T planes. Density, temperature and pressure evolve on the dashed lines from left to right. Solid and thick lines correspond to the thermally stable equilibrium. Dotted lines correspond to the thermally unstable equilibrium. (b) Shaded area denotes thermally unstable region determined by the linear analysis. This figure shows how thermally stable CNM becomes unstable.

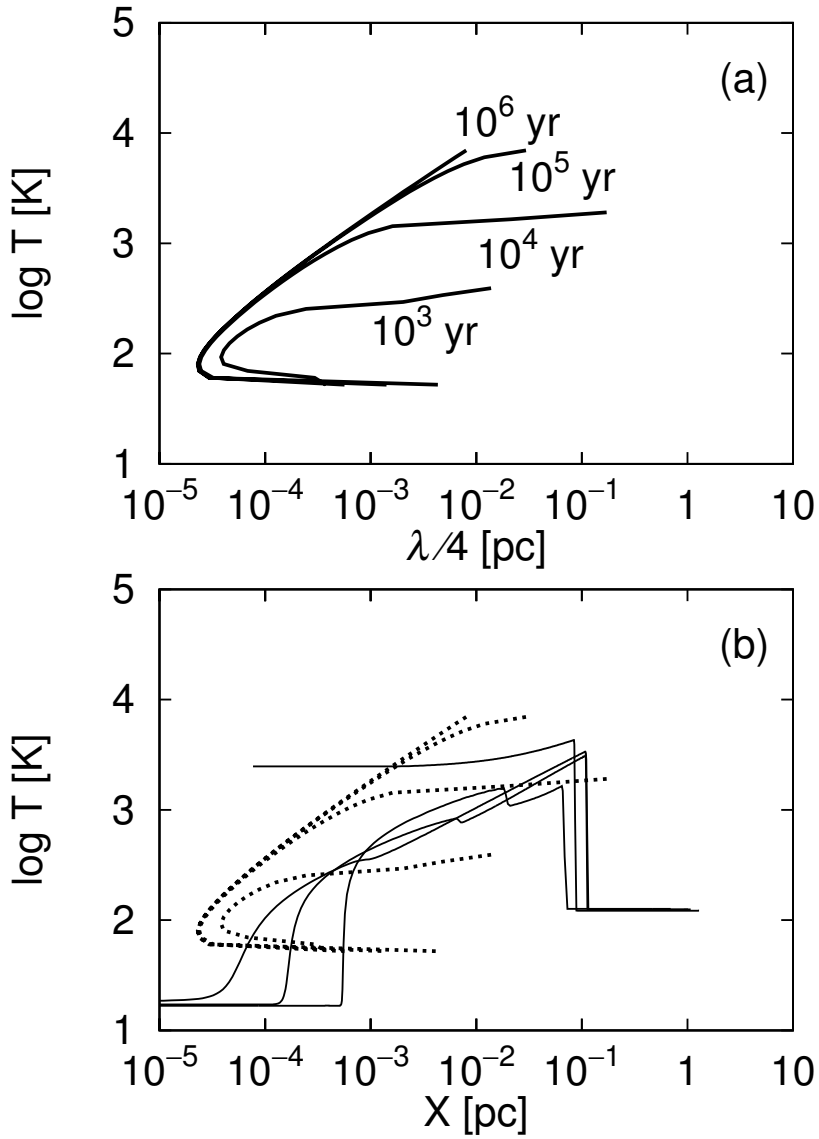


Figure 4.9: Growth time of the instability. The vertical axis denotes unperturbed temperature. The value of the constant pressure, $P_c/k_B = 2 \times 10^5\,\mathrm{K/cm^3}$, is adopted from the result of our nonlinear calculation. The density can be deduced from the relation, $n = P_c/k_B T$. The horizontal axis denotes a quarter of perturbation wavelength. Contours of growth time are depicted in this wavelength-temperature plane. The shortest wavelength of unstable perturbation is $\lambda_{\rm min} \sim 2.5 \times 10^{-5}\,\mathrm{pc} \approx 5\mathrm{AU}$. (b) We superpose the temperature evolution of our nonlinear calculation (Fig. 4.7b) upon the growth rate contours shown in panel a.

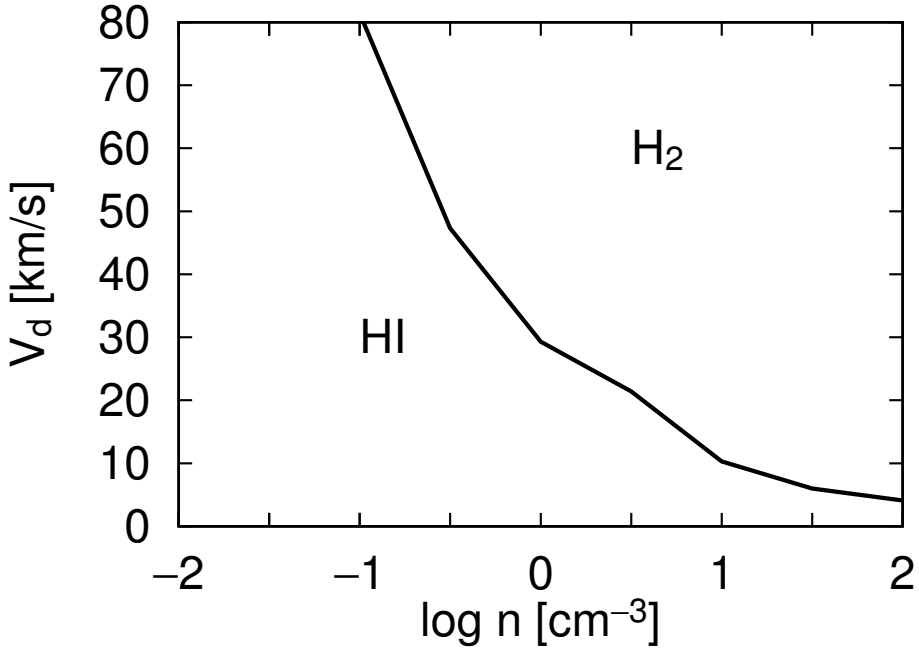


Figure 4.10: Chemical abundance of the thermally collapsed layers in the one-dimensional hydrodynamic calculation. The horizontal axis denotes the initial number density. The vertical axis denotes the velocity difference. Solid lines denote the 8 % boundary of H_2 number fraction.

5

FRAGMENTATION OF THERMALLY COLLAPSED LAYERS

In this chapter, we show that the fragmentation of the shock-compressed layer indeed provides turbulent condensations in cold clouds, by using two-dimensional hydrodynamic calculation with radiative cooling and heating and thermal conduction.

5.1 NUMERICAL SIMULATIONS

5.1.1 Numerical Scheme

The hydrodynamics module of our scheme is based on the second-order Godunov method (van Leer 1997). We solve following hydrodynamic equations.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \tag{5.1}$$

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = -\frac{\nabla P}{\rho}, \tag{5.2}$$

$$\frac{\partial \rho e}{\partial t} + \nabla \cdot (\rho e \mathbf{v}) + P \nabla \cdot \mathbf{v} = \rho\Gamma - \rho^2 \Lambda(T) + \nabla \cdot (K \nabla T), \tag{5.3}$$

where ρ, P, $\mathbf{v}$ are the density, pressure, and velocity of the gas, the specific internal energy $e = P/(\gamma - 1)\rho$, with $\gamma = 5/3$. Γ and Λ are heating and cooling rate, respectively. For the coefficient of thermal conductivity we adopt $K = 2.5 \times 10^3 T^{1/2} \mathrm{cm}^{-1}\mathrm{K}^{-1}\mathrm{s}^{-1}$ (Parker 1953). Effects of self-gravity and magnetic filed are not treated in this paper.

In Chapter 2, we included the following processes in the hydrodynamic calculations: photoelectric heating from small grains and PAHs, heating and ionization by cosmic rays and X-rays, heating by H_2 formation and destruction, atomic line cooling from hydrogen Lyα, C II, O I, Fe II, and Si II, rovibrational line cooling from H_2 and CO, and atomic and molecular collisions with grains. However the following analytic fitting function adopted here reproduces the features relevant for our purpose:

$$\Lambda(T)/\Gamma \quad = \quad 10^7 \exp\left(-114800/(T+1)\right)$$

$$+14\sqrt{T}\exp\left(-92/T\right)\mathrm{cm}^3, \tag{5.4}$$

$$\Gamma = 2\times10^{-26}\mathrm{ergs/s}. \tag{5.5}$$

We adopt linearized implicit time integration for the cooling and heating term.

Thermal conduction could stabilize and even erase a density perturbation if its length scale is smaller than the Field length, $\lambda_{\rm F}=\sqrt{KT/\rho^2\Lambda}$ (Field 1965). The Field length also determines the width of intermediate region (thermal front) between the cold clumpy medium and surrounding warm medium where heating and cooling by the thermal conduction balance the radiative heating and cooling. Note that calculations with large grid spacing cannot resolve thermal front and grids on the surface of cold clumps become numerically unstable. Thus the spatially high resolution is required for the calculation of thermal instability. We use 1024 × 1024 Cartesian grid points covering a 1.2 × 1.2 pc region so that the spatial resolution is 0.00117 pc = 242 AU.

5.1.2 Initial and Boundary Conditions

To investigate the shock propagation into WNM we consider a plane-parallel shocked layer of gas. We take the y-axis to be perpendicular to the layer. Uniform flow approaches the layer from upper side with a velocity $V_{\rm y}=-28.12$ km/s and density $n=0.6$ cm^{-3} and temperature $T=6000$ K. We assume that the lower side of the shocked layer is occupied by hot, tenuous gas (density $n=0.6$ cm^{-3}) with a high pressure $P/k_{\rm B}=4\times10^4$Kcm^{-3}.

We set up density fluctuations as follows:

$$\frac{\delta\rho(x,y)}{\rho_0}=\frac{0.05}{k_{\rm max}}\sum_{k_{\rm x},k_{\rm y}=1}^{k_{\rm max}}\sin(2\pi(xk_{\rm x}+yk_{\rm y})/L+\theta(k_{\rm x},k_{\rm y})), \tag{5.6}$$

where θ is a random phase and $k_{\rm max}=8$. We set out-going boundary condition for the y-direction, and periodic boundary condition for the x- direction.

5.1.3 Results of Thermal Instability and Fragmentation

Figure 5.1, 5.2, 5.3, 5.4 show density distribution at t =0.413, 0.807, 1.474, 1.808 Myr, respectively. Density discontinuity at $y\sim0.4$ pc corresponds to the shock front. Density discontinuity at $y\sim-0.1$ pc is the contact surface between the shock-compressed layer and the hot interior. Thermally collapsed layer and its fragments are seen in the shock-compressed layers.

The upper left panel of the Figure 5.1 shows the beginning of the formation of cold thermally collapsed layer. Behind the shock front, cooling dominates heating and temperature decreases monotonically. The ISM in the range of 300 – 6000 K is thermally unstable. Thus, thermally collapsing layer is subject to thermal instability. The condensations continue to collapse and cool until the radiative cooling balances the radiative heating, in such a way that

approximate pressure equilibrium is maintained between condensations and the surrounding gas. Temperature of these clumps finally reaches the value of thermal equilibrium ($T = 20$ K at $n = 2000\mathrm{cm}^{-3}$). When the cooling balances the heating, the condensation ceases to collapse but the surrounding warm gas continues to cool radiatively and accrete onto the cloud cores.

The cold clumps have considerable translational velocity dispersion. The typical velocity dispersion is about several km/s. The linear analysis predicts that perturbation with sufficiently long wavelength grows exponentially. However, the velocity of the non-linearly developed perturbation has the upper limit that is the sound speed of the warmer medium ($\approx$ 10km/s), because the driving force of the instability is the pressure of the less dense warmer medium. This non-linearly developed perturbation produces the translational velocities of the cold clumps. These velocity on the order of the sound speed of the WNM is highly supersonic with respect to the sound speed of cold gas. Thus we can understand why the supersonic velocity dispersion of the cold medium is comparable to the sound speed of the WNM.

The cold cloud cools the interface via thermal conduction. The pressure of the warm medium between two cold clumps becomes smaller, which in effect provides an attractive force of two adjacent cold clumps. This attractive force is the driving force of the coalescence of two cold clumps. Thus, cold clumps coalesce into larger clouds.

The size of these clumps is a few orders of magnitude smaller than the Jeans length, $\lambda_{\mathrm{J}} = 1.2\ \mathrm{pc}\sqrt{(T/20\mathrm{K})(2000\mathrm{cm}^{-3}/n)}$. Thus, each cloud is gravitationally stable. The shock front itself remains stable in 10^6 years in this plane-parallel problem. From these results we demonstrate that the role of thermal instability behind the shock front is one of the most important processes in the evolution of interstellar shocked layers.

5.2 DISCUSSION

5.2.1 Formation of Molecules

The main coolant in cold dense clouds is supposed to be CO (see, e.g., Neufeld, Leep, & Melnick 1995; Paper I). H_2 enables the formation of CO and other molecules. The formation time of H_2 in the CNM critically depends on the optical depth and local density. The formation time of H_2 on grain surface is $(Rn_{\mathrm{HI}})^{-1} = 1.5(2000\mathrm{cm}^{-3}/n)$ Myr (see Fig. 4.1d). The column density of the clump becomes $\Sigma = 0.72 \times 10^{21}(n/2000\mathrm{cm}^{-3})(R/0.08\mathrm{pc})\mathrm{cm}^{-2}$. Therefore, the condition of efficient molecule formation is satisfied. We calculated detailed thermal-chemical equilibrium state in the dense region by adopting the values of density, temperature, and column density from this dynamical simulation. All of the hydrogen is H_2 and 5.8% of the carbon is CO in thermal and chemical equilibrium.

We simulate the observation of molecular clumps at $t = 1.808$ Myr (Fig. 5.4) as the Position-Velocity (P-V) diagram of the ^{12}CO J = 1 – 0 emission (upper panel of Figure 5.5).

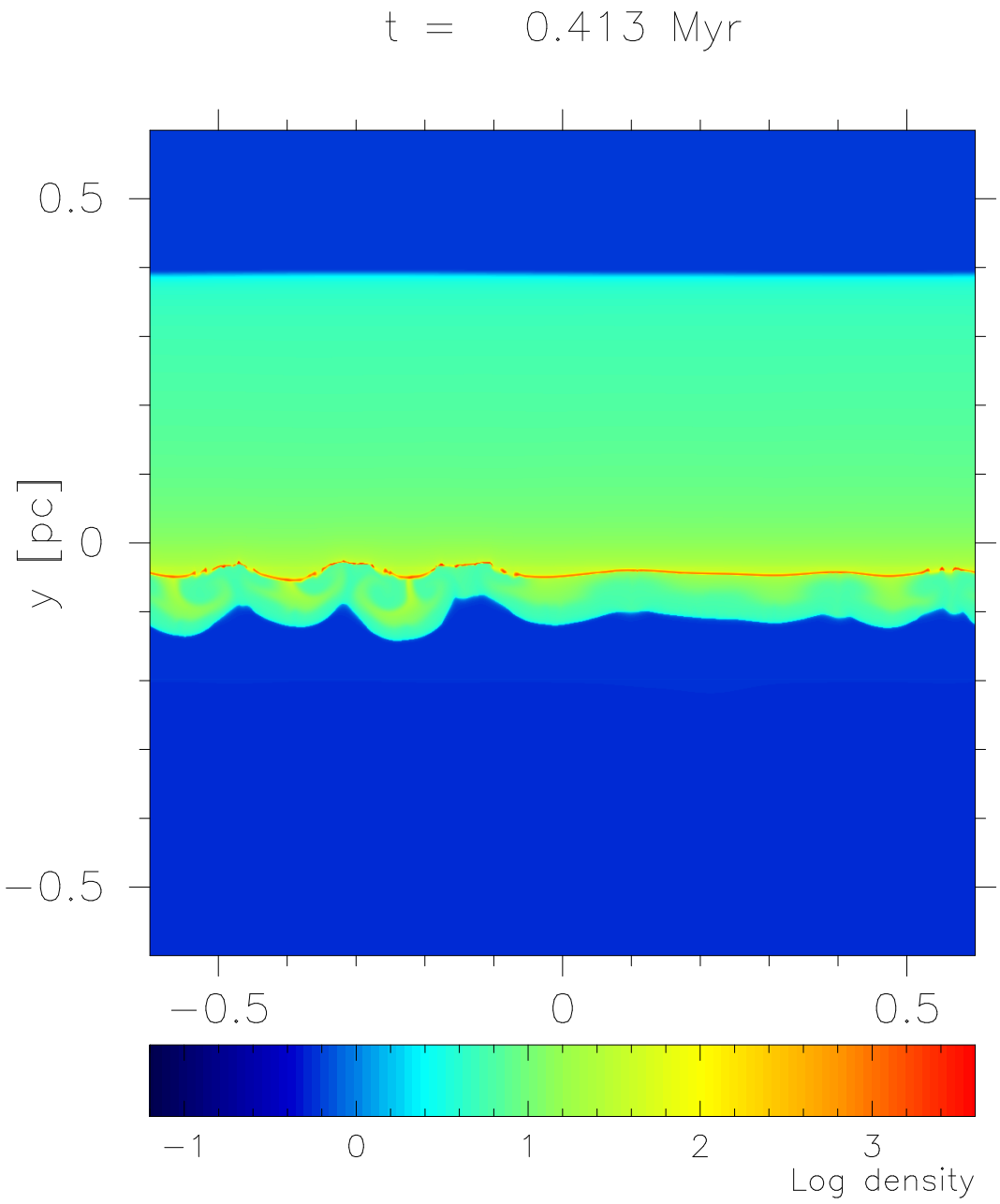


Figure 5.1: Density distribution of the shock-compressed layers. The panel shows the calculation at $t = 0.413$ Myr.

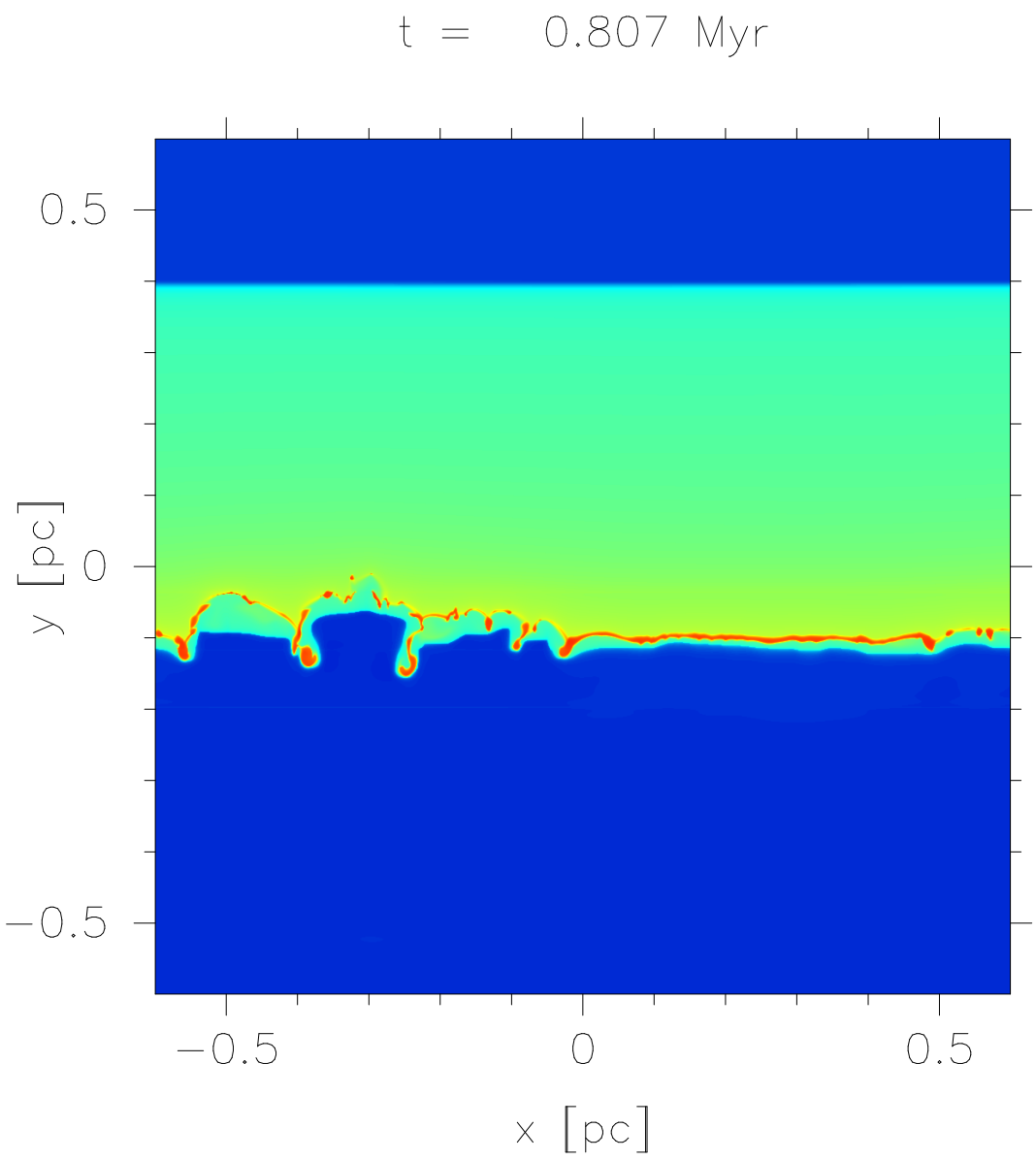


Figure 5.2: Same as Figure 5.1, but at $t = 0.807$ Myr.

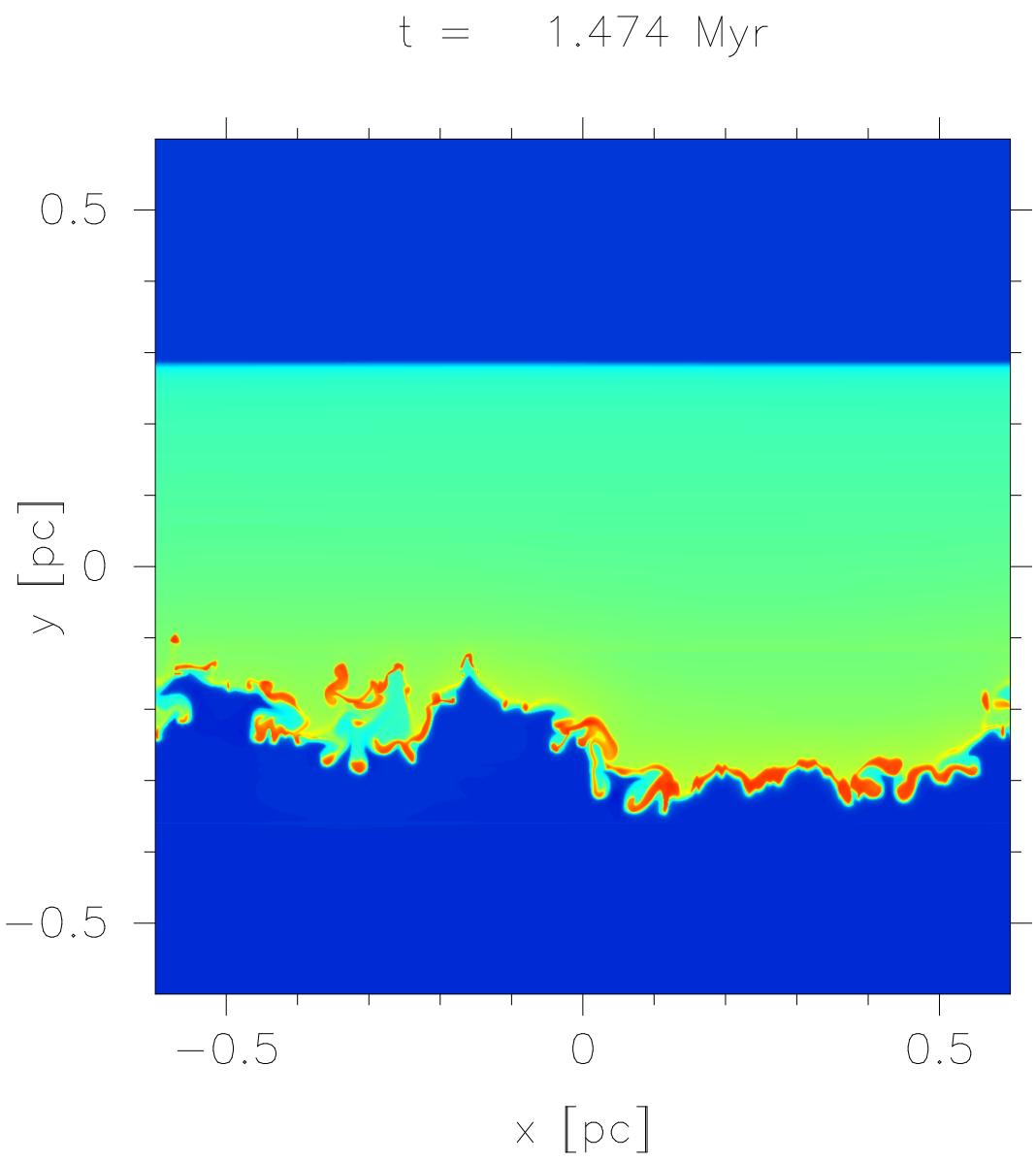


Figure 5.3: Same as Figure 5.1, but at $t = 0.474$ Myr.

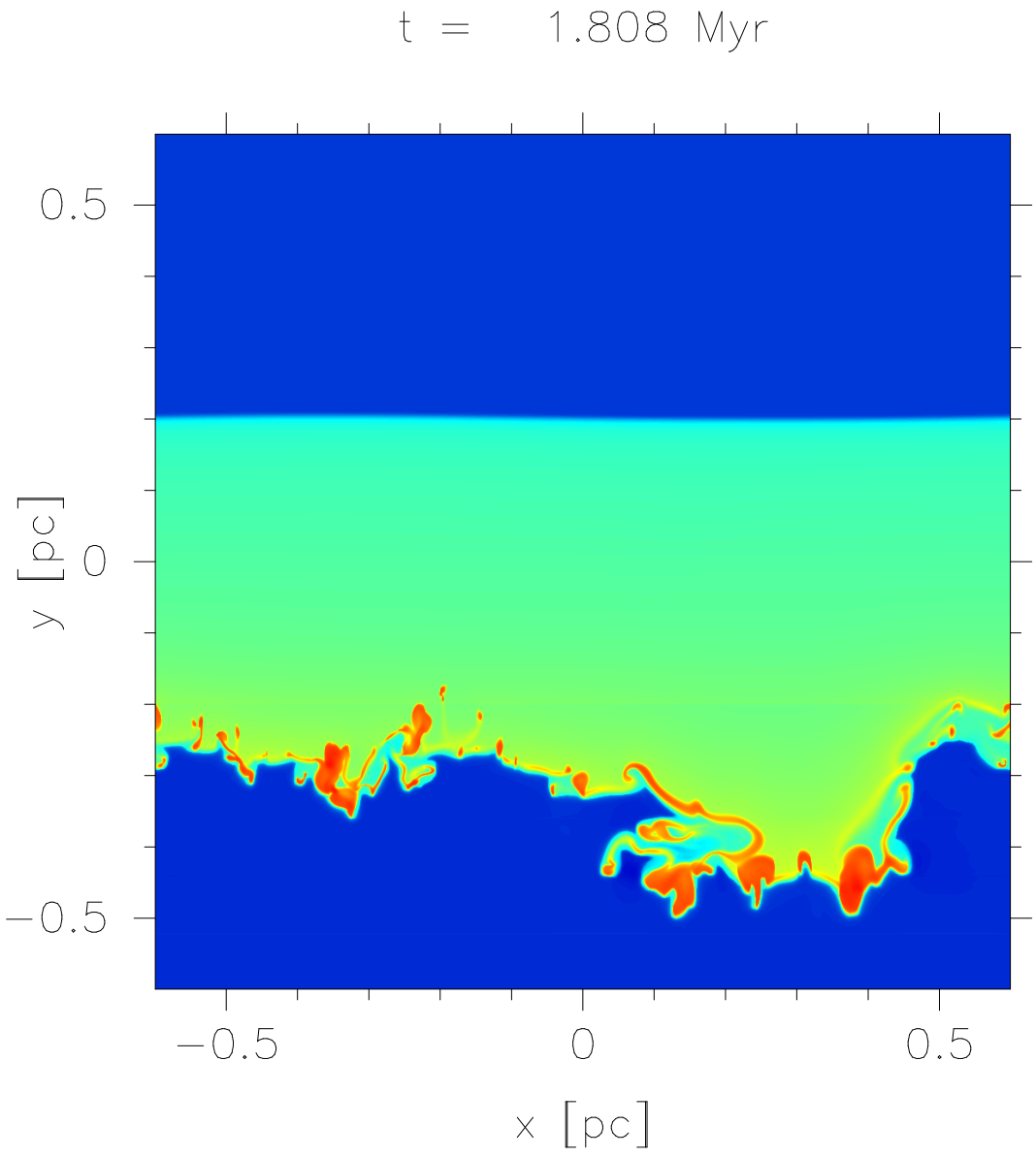


Figure 5.4: Same as Figure 5.1, but at $t = 1.808$ Myr.

We use the usual relationship $T_{\rm B} = \frac{1}{2}\lambda^2 I/k_{\rm B}$, with the specific intensity $I = n^2\Lambda L/(4\pi\Delta v)$, where $n^2\Lambda$ is the cooling per unit volume, L is the path length, Δv is the Doppler line width. We adopt Δv to be the sound velocity. As shown in Figure 5.5, the typical velocity dispersion in the clumps is about a few km/s. The cloud to cloud velocity dispersion is also a few km/s. The lower panel shows that the P-V diagram of hydrogen nuclei. The velocity dispersion of the diffuse warm medium is about 10 km/s.

5.2.2 An Origin of "Turbulence" in Interstellar Clouds

Diffuse ISM in the Galaxy is frequently compressed by supernova explosions (McKee & Ostriker 1977). Therefore, the shock propagation into the ISM plays an important role in the evolution of the Galactic ISM. We have studied the shock propagation into WNM by two-dimensional hydrodynamic calculations, and have shown that the thermally collapsed layer breaks up into very small cloudlets. Fragmentation of the thermally collapsing layer is a result of thermal instability. Thermal instability produces many cloudlets which have different translational velocities. We expect that the Galactic ISM is occupied by the small cloudlets which have supersonic velocity dispersion , because the ISM is frequently compressed by supernova explosions, stellar winds, spiral density waves, cloud-cloud collisions, etc. The H I 21-cm observation maps show the existence of many shell-like or filamentary structures in the Galaxy (Hartmann & Burton 1997). Actually, high resolution observations show that the clumpy distribution is ubiquitous in the Galaxy (Heiles 1997). If these structures are really the results of the shock waves, many small molecular cloudlets should be hidden in the shock-compressed layers. These cloudlets should have translational velocities owing to the fragmentation of thermal instability. Observational "Turbulence" in the ISM should reflect these motions. Thus, we propose that an origin of "Turbulence" in the ISM is the motion of small cloudlet complex.

In the radiative shocked layers, the gases lose thermal energy through radiative cooling. Thus, the initial kinetic energy of the pre-shock gas (in the comoving frame of the postshock gas) converted to the radiation energy, which escapes from the system. If, however, the postshock gas becomes dynamically unstable by the thermal instability as in this chapter, some portion of the thermal energy is transformed into the kinetic energy of the translational motions of the cold cloudlets, which does not easily escape from the system. Therefore we can consider the origin of interstellar turbulence is due to the conversion of the gas energy in supersonic motion. In principle, the kinetic energy of the cloudlets can be lost via coalescence of the cloudlets. The damping rate of the velocity dispersion of the cloudlets will play a key role in the evolution of this system, and hence, should be studied in the subsequent paper.

Expected effects of magnetic field modify the development of the thermal instability described here. The presence of magnetic field can suppress thermal conduction efficiently, allowing the collapse of small scale structure by thermal instability. The magnetic forces suppress the motion that is perpendicular to the magnetic field lines, and hence suppress the growth of perturbations whose wavevector is perpendicular to the magnetic field lines.

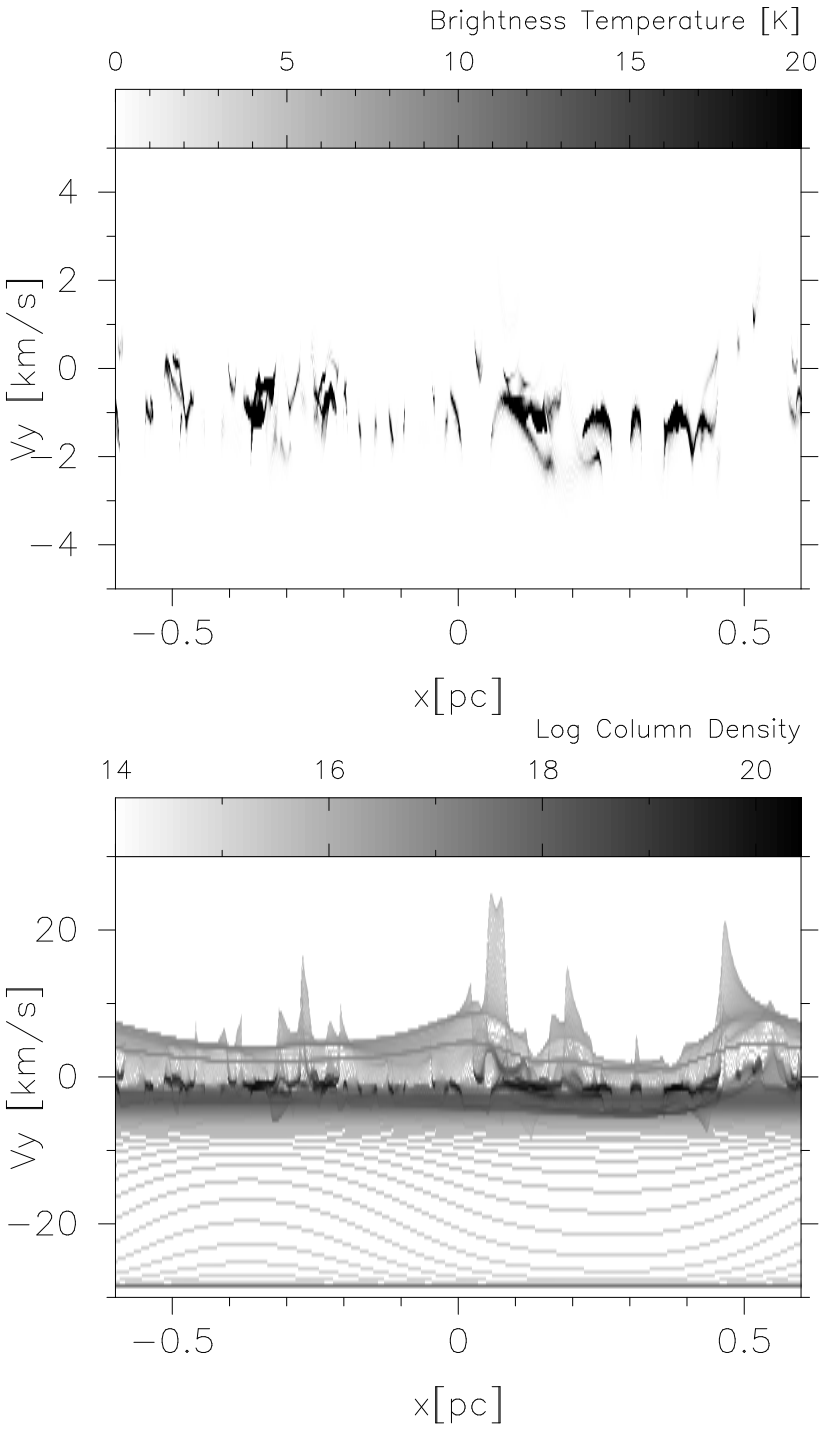


Figure 5.5: The upper panel shows the position-velocity (P-V) diagram of the ^{12}CO J=1 – 0 emission obtained from the calculation at $t = 1.808$ Myr (see Fig. 5.4). We assume 5.8 % of the carbon is CO. The lower panel shows the P-V diagram of the hydrogen nuclei.

However, the magnetic forces do not affect the motion that is parallel to the magnetic field lines, and hence the growth of perturbations whose wavevector is parallel to the magnetic field lines. As a result, the dense sheets or filaments will form and tend to align *perpendicular* to the magnetic field lines. We need three-dimensional calculation to analyze these effects of magnetic fields.

6

EVOLUTION OF INTERSTELLAR COLD CLOUDS

We investigate the evolution of cold clumps in a warm medium by one-dimensional hydrodynamic calculations with treatments of heating, cooling, and thermal conductivity by contrasting the difference with the previous semi-analytic works that include classical works of Zel'dovich & Pikel'ner (1969) and Penston & Brown (1970).

This chapter is organized as follows. In §6.1, we explain our numerical calculations. In §6.2, we present the results of our calculations of two phase medium. In §6.3, we summarize the results and disscuss the implications.

6.1 NUMERICAL METHODS

6.1.1 Hydrodynamic Equations

To study the realistic thermal and dynamical evolution of two phase medium, we set up a two-phase medium via thermal instability and follow the long term evolution of the medium. We solve the following hydrodynamic equations:

$$\frac{d\rho}{dt} + \rho\frac{\partial v}{\partial X} = 0, \tag{6.1}$$

$$\rho\frac{dv}{dt} = -\frac{\partial P}{\partial X}, \tag{6.2}$$

$$P = \frac{\rho k_{\rm B} T}{m_{\rm H}}, \tag{6.3}$$

$$\frac{1}{\gamma-1}\frac{dP}{dt} - \frac{\gamma}{\gamma-1}\frac{P}{\rho}\frac{d\rho}{dt} = -\rho\mathcal{L} + \frac{\partial}{\partial X}K\frac{\partial T}{\partial X}. \tag{6.4}$$

In the above equations, ρ is the mass density of the gas, v is the velocity of fluid elements, P is the gas pressure, and K is the coefficient of thermal conductivity. For the range of temperatures considered, the dominant contribution to thermal conductivity is that due to neutral atoms, for which $K = 2.5 \times 10^3 T^{1/2}\,{\rm cm^{-1}K^{-1}s^{-1}}$ (Parker 1953). We use the ratio of the specific heats $\gamma = 5/3$ for simplicity. For the heating/cooling function, we adopt

$\rho\mathcal{L} = -\Gamma n + \Lambda(T)n^2$ with $\Gamma = 2 \times 10^{-26}$ergs/s and for Λ, an following analytical fit of the cooling function,

$$\begin{aligned} \Lambda(T)/\Gamma &= 10^7 \exp\left(-114800/(T+1)\right) \\ &\quad +14\sqrt{T}\exp\left(-92/T\right), \end{aligned} \tag{6.5}$$

calculated in Koyama & Inutsuka (2000; Chapter 2), including photoelectric heating from small grains and PAHs, heating and ionization by cosmic rays and X-rays, heating by H_2 formation and destruction, atomic line cooling from hydrogen Lyα, C II, O I, Fe II, and Si II, rovibrational line cooling from H_2 and CO, and atomic and molecular collisions with grains. We only consider optically thin cooling function because the column density of the cloud is sufficiently small.

The hydrodynamics module of our scheme is based on the second-order Godunov method. We use the result of Riemann problem at each cell interface to calculate numerical fluxes, without introducing any artificial viscosity, which characterizes the Godunov method. Higher-order Godunov method has been the state-of-the-art numerical method in last two decades (see e.g., van Leer 1997), and widely used in astrophysical context (see e.g., Truelove et al. 1998).

6.1.2 Initial and Boundary condition

The initial condition is a static unstable equilibrium perturbed by the linear eigenfunction of the unstable mode. The typical models are summarized in Table 6.1. The length of the computational domain corresponds to two wavelengths of the eigenfunction. Initial temperatures are taken to be the value of thermal equilibrium state (see Fig. 6.1a). We have follow the evolution until $t = 50 - -100$ Myr. The boundary conditions are periodic boundary conditions, i.e., the length of the computational domain is fixed. Note that the thermal conductive flow only changes the spatial distribution of thermal energy within the system. while the radiative heating and cooling changes the total energy of the system.

6.2 RESULTS

Figure 6.1a shows the equilibrium pressure as functions of density. Negative slope corresponds to thermally unstable equilibrium. Initial conditions of models are ploted on figure 6.1a (*circle*). Field (1965) analyzed the local linear stability of uniform gas in thermal equilibrium and found the dispersion relation as

$$\begin{aligned} &\left[\sigma^2 + (kc_{\rm s})^2\right]\left(\sigma + \frac{s_{\rm T}}{\tau_{\rm cool}} + \ell c_{\rm s} k^2\right) \\ &\quad -(kc_{\rm s})^2\left[(1-\gamma)\sigma + \frac{1}{\tau_{\rm cool}}\right] = 0, \end{aligned} \tag{6.6}$$

where

$$s_{\rm T} = \partial \ln \Lambda / \partial \ln T, \tag{6.7}$$

$$\tau_{\rm cool}^{-1} = (\gamma - 1)\Lambda / k_{\rm B} T, \tag{6.8}$$

$$c_{\rm s} = \sqrt{\gamma P / \rho}. \tag{6.9}$$

We adopt the collision mean free path, ℓ, instead of thermal conduction coefficient $K = nk_{\rm B}\ell c_{\rm s}/(\gamma - 1)$. The dispersion relation of each model is shown in Figure 6.1 b.

6.2.1 Linear growth of thermally unstable medium

In this subsection, we take model D as a typical model. Linear growth rate of the instability in this model is $1.46 \times 10^{-6}{\rm yr}^{-1}$. The perturbation grows as $\exp(1.46 \times 10^{-6}t)$. We set eigenfunction of the growing mode with the amplitude $\delta\rho/\rho_0 = 0.01$ and wavelength $2\pi/k =$ 0.6pc. Figure 6.1 d shows the evolution of the fluid element which has the maximum density at each time step (*solid line*). The agreement between the numerical results (*solid line*) and analytical linear growth solution (*dashed line*) is excellent.

Around $t = 2.9 \times 10^6$yr, density growth was satulated because density attained the thermally equilibrium value.

6.2.2 Quasi-steady evolution of two phase medium

In this subsection, evolution after cloud collapse is shown. Thermal timescale of overdensity region decrease with time because density increase with time. On the other hand, thermal timescale of underdensity region increase with time. Thus the evolution after the collapse of the cold regions should be analysed in terms of slower evolution of warm intercloud gases (see Fig. 6.2a). Figure 6.2b shows the velocity. Intercloud gases expand and accrete into the clouds. The gas pressure is uniformly distributed through cloud-intercloud system but varies with time (Fig. 6.2c). Figure 6.2d shows the mass flux. ZP and PB estimated the mass flux by the quasistatic approximation. Their analyses are based on the following assumptions:

1. The cloud is assumed to change so slowly in time that they may allow a quasistatic approximation by neglecting the Eulerian time derivatives in the hydrodynamic equations (6.1,6.2,6.4). Therefore the equation (6.1) gives $\rho v =$ const.

2. They assume that the pressure does not change appreciably when passing through the cloud surface. They therefore put $P =$ const.

Then the energy equation (6.4) becomes,

$$\rho v \frac{\partial}{\partial X}\left(\frac{\gamma}{\gamma - 1}\frac{P}{\rho}\right) = -\rho\mathcal{L} + \frac{\partial}{\partial X} K \frac{\partial T}{\partial X}, \tag{6.10}$$

where left hand sides is mass flux times Eulerian space derivative of the enthalpy per unit mass. Now at both sides of this interface are the equilibrium states of cold cloud and warm intercloud, therefore in these regions no conduction occurs and we may write down the boundary conditions of equation (6.10) as the fact there is no temperature gradient in these regions. The physical significance of equation (6.10) is more clearly seen if we take its formal first integral:

$$\rho v \left(H_{\rm w} - H_{\rm c}\right) = -\int \rho \mathcal{L}\, dX, \tag{6.11}$$

where $H_{\rm w}, H_{\rm c}$ are enthalpy of intercloud and cloud respectively. The condition that no condensation or evaporation take place is $\rho v = 0$. When the right hand side of the integral does not vanish, one phase will transform into other with a finite rate, corresponding to a mass flux. Then they found that the mass flux was the order of 100 $m_{\rm H}$ cm^{-2}s^{-1} and the timescale for condensation or evaporation of one solar mass was 10^9 year. They showed that clouds evapolate or condense only on a very long timescale. Thus they concluded that the true equilibrium of clouds and intercloud medium at saturation pressure of the interstellar medium is never attained. The result of our dynamical calculation sharply contradicts their prediction.

Our hydrodynamic calculation shows that the accretion of the warm gas onto the cold cloud is a dynamical process. Most importantly, we find that the pressure is no longer constant with time (see Fig. 6.2c and 6.4b). This means that the time derivatives should not be neglected. Figure 6.3 plots the snapshots of the evolution of two phase system in the $\rho - P$ plane. Labels 1 and 2 are the snapshots corresponding to the dashed line and dotted line in Figure 6.1c, respectively. Labels 3,4, and 5 correspond to the solid line, the dashed line, and the dotted line in Figure 6.2, respectively. The most important difference of our hydrodynamic calculations from the preceding hydrostatic analyses is the existence of time-variability of the pressure. In the next subsection, we present a better analytical approximation for two phase medium.

6.2.3 Energy reconstruction

Figure 6.4a shows the time evolution of the fluid element which has the maximum density (i.e., cold cloud density;*solid line*) and minimum density (i.e., intercloud density;*dashed line*). We distinguish three stages of the evolution. Stage I corresponds to the linear growth stage. Perturbation quickly grow and forms cold clouds. Stage II corresponds to the stage that the heating in the warm intercloud gas dominates. At the end of this stage, the warm intercloud region attains thermal equilibrium temperature. In stage III the system is approaching to the saturated pressure.

Figure 6.4b shows the time evolution of the fluid element which has the maxmun pressure (*solid line*) and minimum pressure (*dashed line*), although two lines coincide in this plot (see also Fig. 6.2c,6.3). Dashed line in Figure 6.4c denotes the mean thermal energy per unit mass. One thousand times of the mean kinetic energy per unit mass is also plotted in figure

6.4c (*solid line*). These figures show that kinetic energy is negligible and the system evolves spatially isobarically. From the above numerical results we assume that the pressure time-variability is one of the most important terms in the left hand side of the equation of energy (6.4). Then the equation of energy (6.4) integrated over space is,

$$-\frac{1}{\gamma-1}\frac{\partial P}{\partial t}L \approx \int \rho\mathcal{L}\, dX, \tag{6.12}$$

where L denotes the computational domain. Figure 6.4d shows the net radiative loss rate per H atom averaged over the system (*dashed line*). Solid line denotes $-\partial P/\partial t/(\gamma-1)$ calculated numerically. The agreement between the net loss rate and the pressure time derivative is excellent. This means that the other term in the left hand side of equation (6.4) is negligible in all stages.

6.2.4 Results from various initial parameters

In this subsection, we summarize the results from several initial parameters. Figure 6.5a denotes the models with the same wavelength (0.6 pc). In this figure calculations from various initial conditions converge to the same pressure evolution. Figure 6.5b denotes time-evlution of the column density of the cloud. All grid elements with a density above 10 cm^{-3} are counted as belonging to the cold cloud. The mass fractions of the cloud in the final time step are tabulated in Table 6.1. The mass fraction depend on the total column density. Figure 6.5c and 6.5d is the temperature and the net loss rate per H atom respectively. All models have similar structures but different sizes of cloud cores. This size of the cloud core depends on the masses in the computational domain. From these results, we find that the clouds with the different sizes formed with the same timescale.

6.2.5 Saturation time

Figure 6.6a denotes the calculations with various initial wavelength (model D: 0.6 pc, D2: 1.2 pc, D3: 2.4 pc, and D4: 4.8 pc). This figure shows that saturation time proportional to the size of the calculation domain. This relationship demonstrates the importance of the sound-crossing timescale. Filled circles denote the time at $t = \alpha L/c_{\mathrm{s}}$ where c_{s}=10km/s is the sound velocity of the warm medium which dominates in volume. From the numerical calculations, we obtain the numerical factor $\alpha \approx 80$.

6.3 DISCUSSION

We have done one-dimensional plane-parallel hydrodynamic calculation to analyze the long term evolution of the interstellar two phase medium. The calculation shows that the pressure is no longer constant with time. We find that the true equilibrium of cold clouds and intercloud warm medium with the saturation pressure of the interstellar medium is attained

within 80 times the sound-crossing timescale in the intercloud warm medium, which is much shorter than the timescale previously believed.

In this study we have revealed basic physics of the interstellar two phase medium. Let us now discuss the possible implications and prepare for future developments.

The mass exchange between the various phases of the ISM plays an important role in galactic evolution. The small H I clouds phase appears to play a crucial role in the transition from diffuse phase to regions of star formation. The fragmentation of dense supernovae remnant shells which have been included in the regulated model of the ISM as the unique mechanism for forming small clouds (McKee & Ostriker 1977). Hennebelle & Pérault (1999) suggested that the linearly stable warm medium is non-linearly unstable: sufficiently large perturbations such as a trans-sonic turbulent WNM may kick the gas to the other stability branch. We have studied the shock propagation into ISM and suggested the important role of thermal instability (Paper I). We have also done two-dimensional hydrodynamic calculations to demonstrate the fragmentation due to thermal instability in shock-compressed layers (Koyama & S.Inutsuka 2001). This calculations shows that the fragments had velocity dispersion in the warm neutral medium and tend to coalesce and become larger clouds.

The mass exchange between the warm medium and the cold medium is strongly dependent on the local ISM conditions. Inutsuka & Koyama (1999) showed that there is a critical column density at which the medium loses its warm phase and starts to coagulate rapidly. The effect due to the change of the ambient radiation field (heating sources) will be key issues in future investigations.

Table 6.1. MODELS.

Model	$n_{\rm i}$[a]	$P_{\rm i}$[b]	$T_{\rm i}$[c]	λ[d]	σ[e]	$P_{\rm f}$[f]	$n_{\rm c}$[g]	$n_{\rm w}$[h]	f[i]
A	1.18	4.54	3853	0.6	0.11	2.83582	51.7815	0.442640	0.453
B	1.51	3.65	2420	0.6	0.41	2.82952	51.5679	0.441420	0.585
C	2.08	2.82	1353	0.6	0.87	2.83628	51.7970	0.442744	0.703
D	2.93	2.22	756	0.6	1.46	2.82903	51.5510	0.441349	0.797
D2	2.93	2.22	756	1.2	1.47	2.82968	51.5727	0.441128	0.824
D3	2.93	2.22	756	2.4	1.42	2.84457	52.0752	0.443814	0.839
D4	2.93	2.22	756	4.8	1.26	2.86042	52.6111	0.446703	0.845
E	4.32	1.83	423	0.6	1.94	2.87537	53.1166	0.449915	0.875

[a]The initial number density in $\rm cm^{-3}$.

[b]The initial pressure in 1000 K $\rm cm^{-3}$.

[c]The initial temperature in K.

[d]The wavelength in pc.

[e]The linear growth rate in 10^{-6} $\rm yr^{-1}$.

[f]The final pressure in 1000 K $\rm cm^{-3}$.

[g]The final number density of the cloud in $\rm cm^{-3}$.

[h]The final number density of the intercloud in $\rm cm^{-3}$.

[i]The cloud column density over total column.

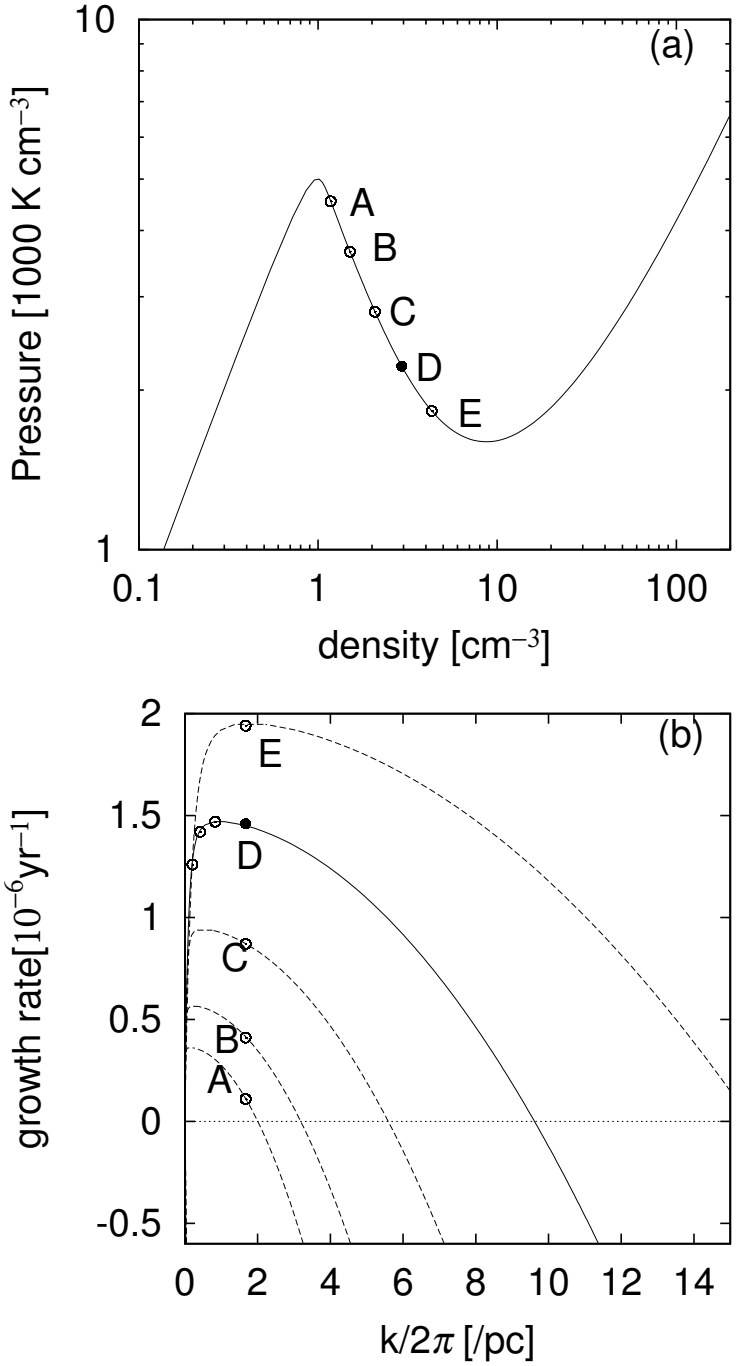
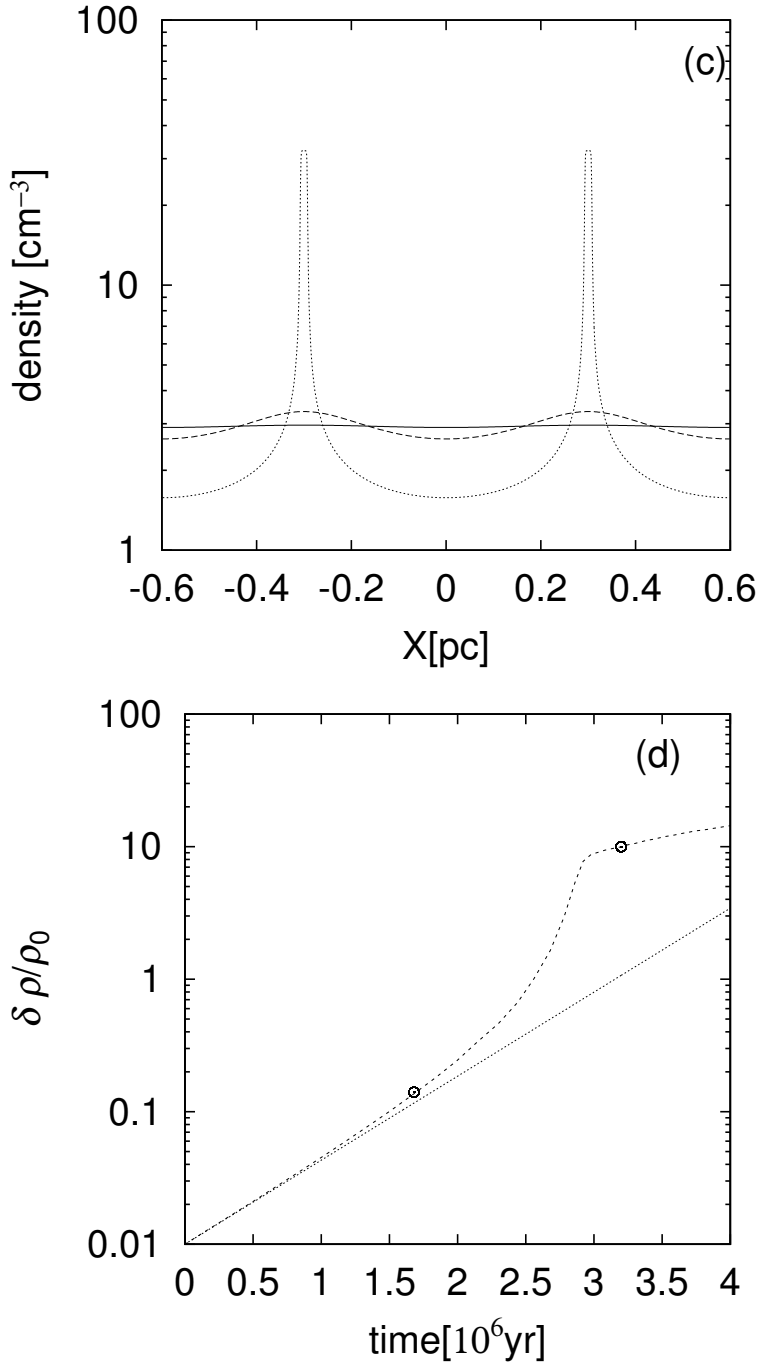


Figure 6.1: (a) Equilibrium pressure. (b) Dispersion relation of the thermal instability. Open circles correspond to the models, and filled circle corresponds to the model D. (c) Hydrodynamic calculation of model D at three different times: 0 yr (*solid line*), 1.68 $\times 10^6$ yr (*dashed line*), 3.2 $\times 10^6$ yr (*dotted line*). (d) Time evolution of density perturbation (*solid line*). Dashed line denotes linear growth. Open circles correspond to the same snapshot in panel c.


100
10
1
density [cm^{-3}]
(c)
-0.6 -0.4 -0.2 0 0.2 0.4 0.6
X[pc]
100
10
1
0.1
0.01
$\delta\rho/\rho_0$
(d)
0 0.5 1 1.5 2 2.5 3 3.5 4
time[10^6yr]

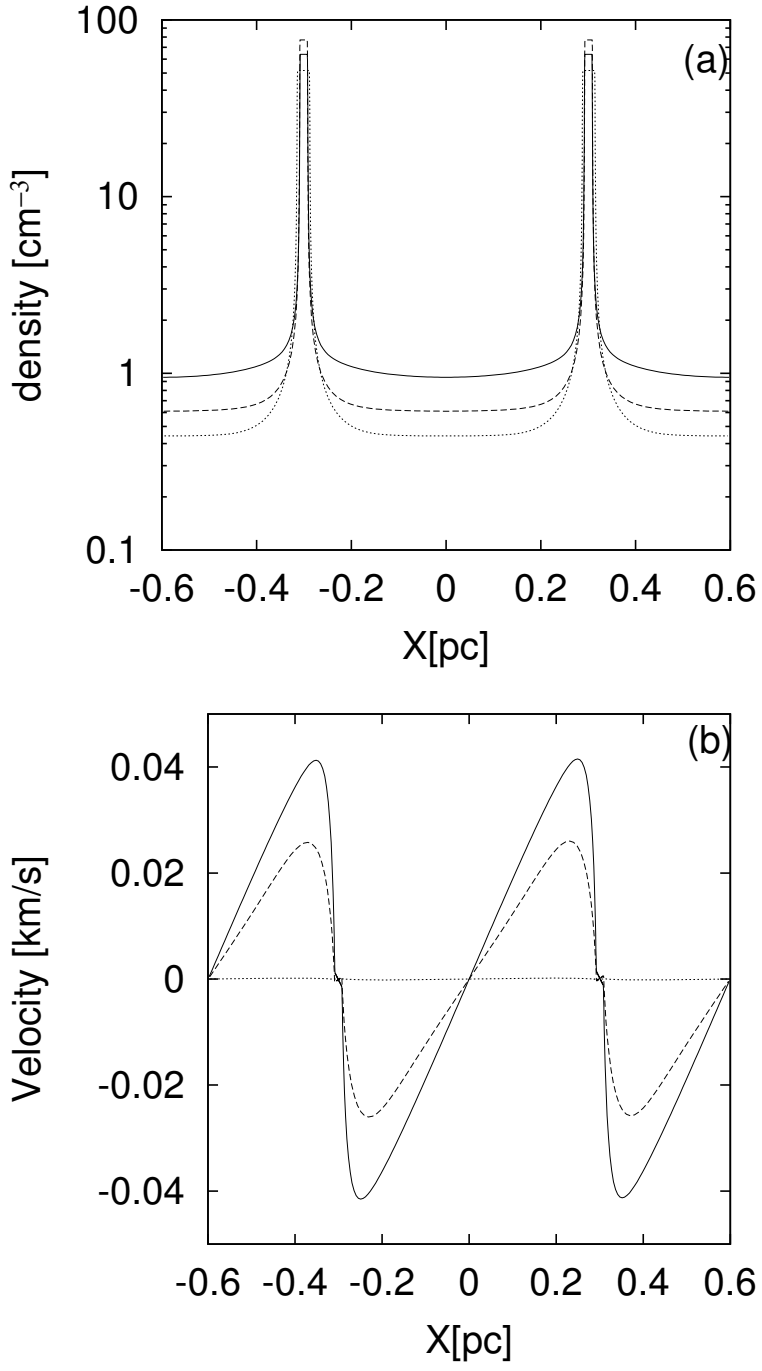
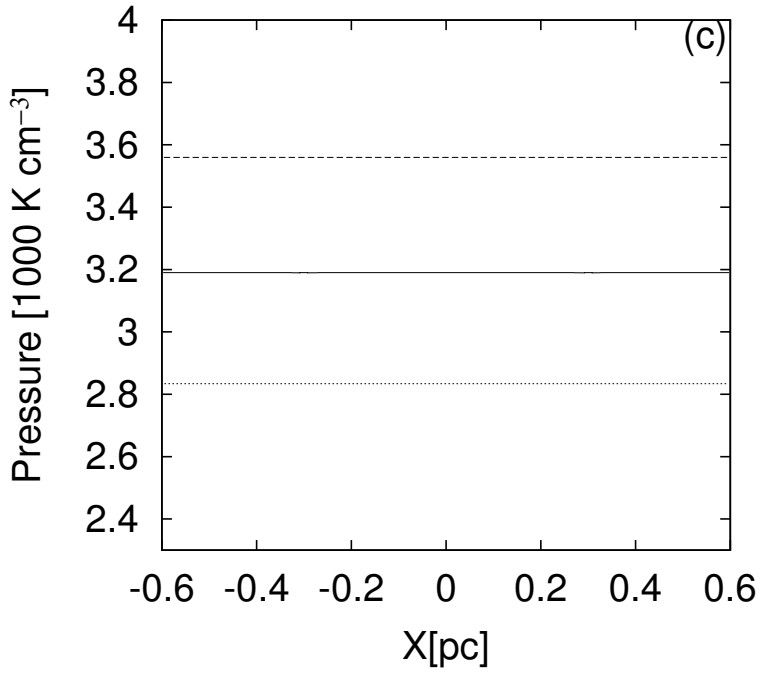
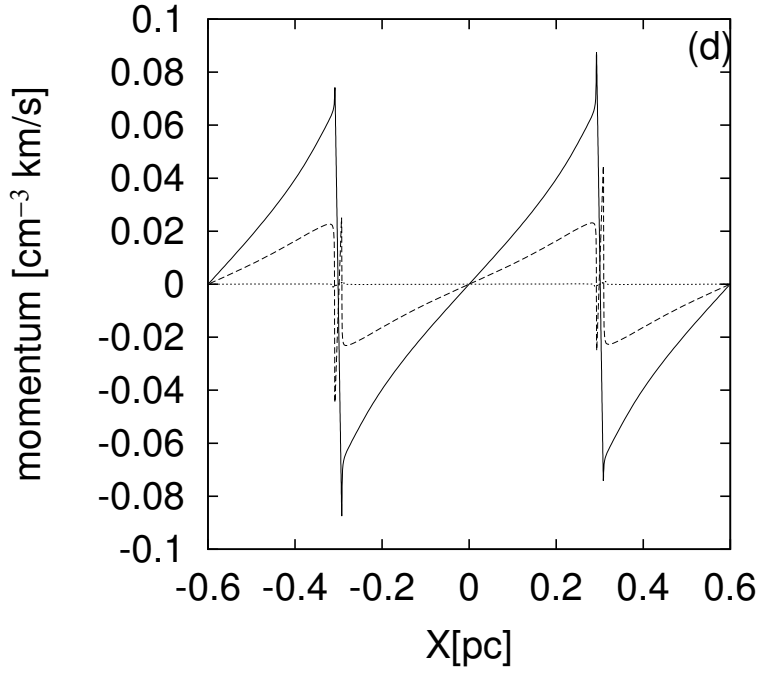


Figure 6.2: The evolution of the two phase medium after the collapse of cold regions. The figures show (a) density, (b) velocity, (c) pressure, (d) mass flux (momentum) which is defined by ρv. Solid lines denote at 5.6 $\times 10^6$ yr, dashed lines at 8.6 $\times 10^6$ yr, dotted lines at 32.2 $\times 10^6$ yr.


(c)
Pressure [1000 K cm^{-3}]
4
3.8
3.6
3.4
3.2
3
2.8
2.6
2.4
-0.6 -0.4 -0.2 0 0.2 0.4 0.6
X[pc]


(d)
momentum [cm^{-3} km/s]
0.1
0.08
0.06
0.04
0.02
0
-0.02
-0.04
-0.06
-0.08
-0.1
-0.6 -0.4 -0.2 0 0.2 0.4 0.6
X[pc]

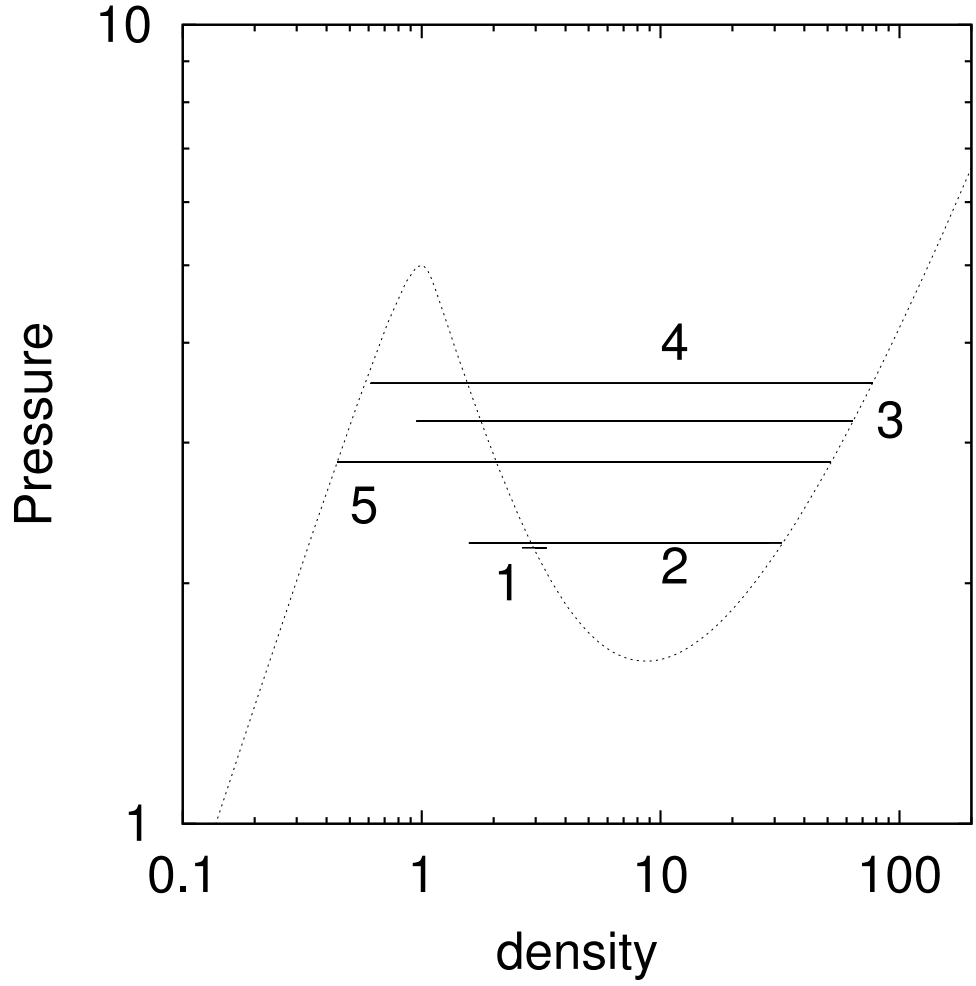


Figure 6.3: Evolution of two phase system in the $\rho - P$ plane. Labels 1 and 2 are the snapshots corresponding to the dashed line and dotted line in Figure 6.1c, respectively. Labels 3,4, and 5 correspond to the solid line, the dashed line, and the dotted line in Figure 6.2, respectively.

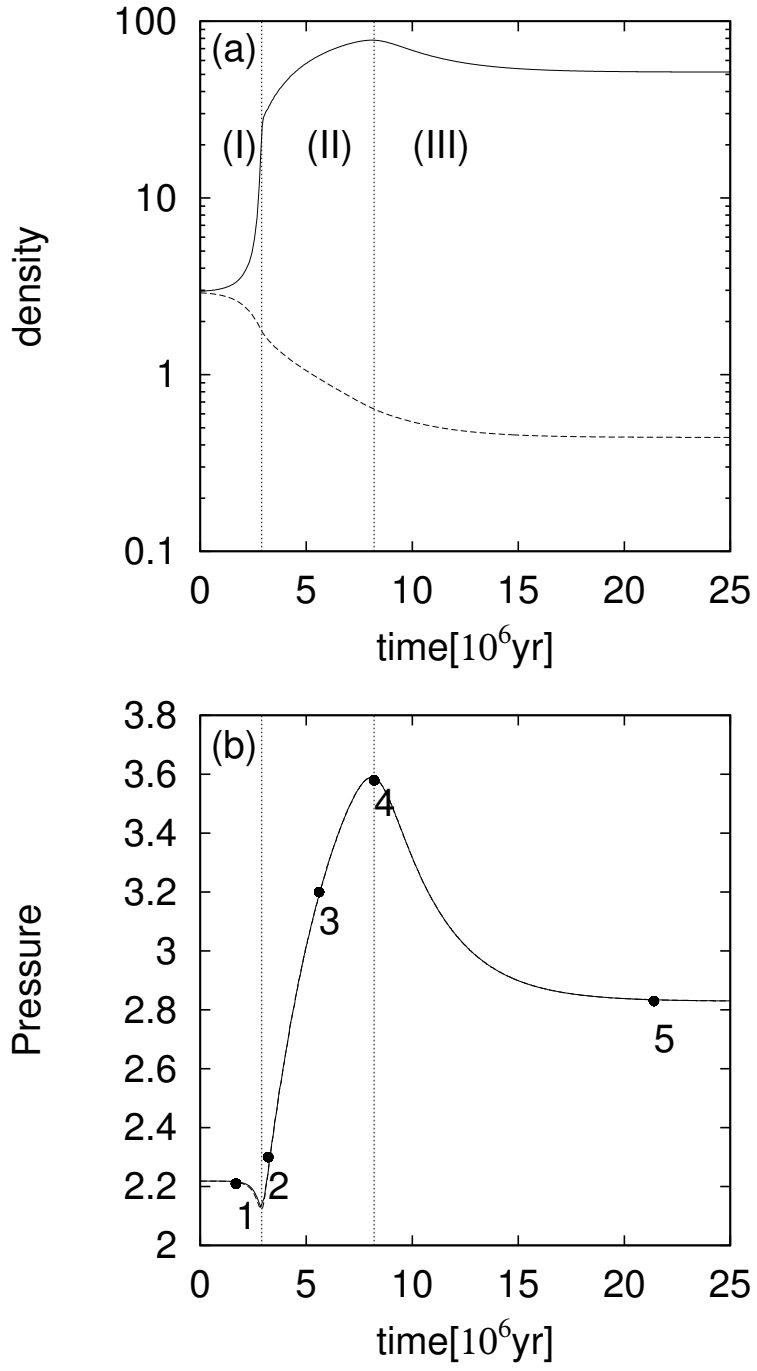
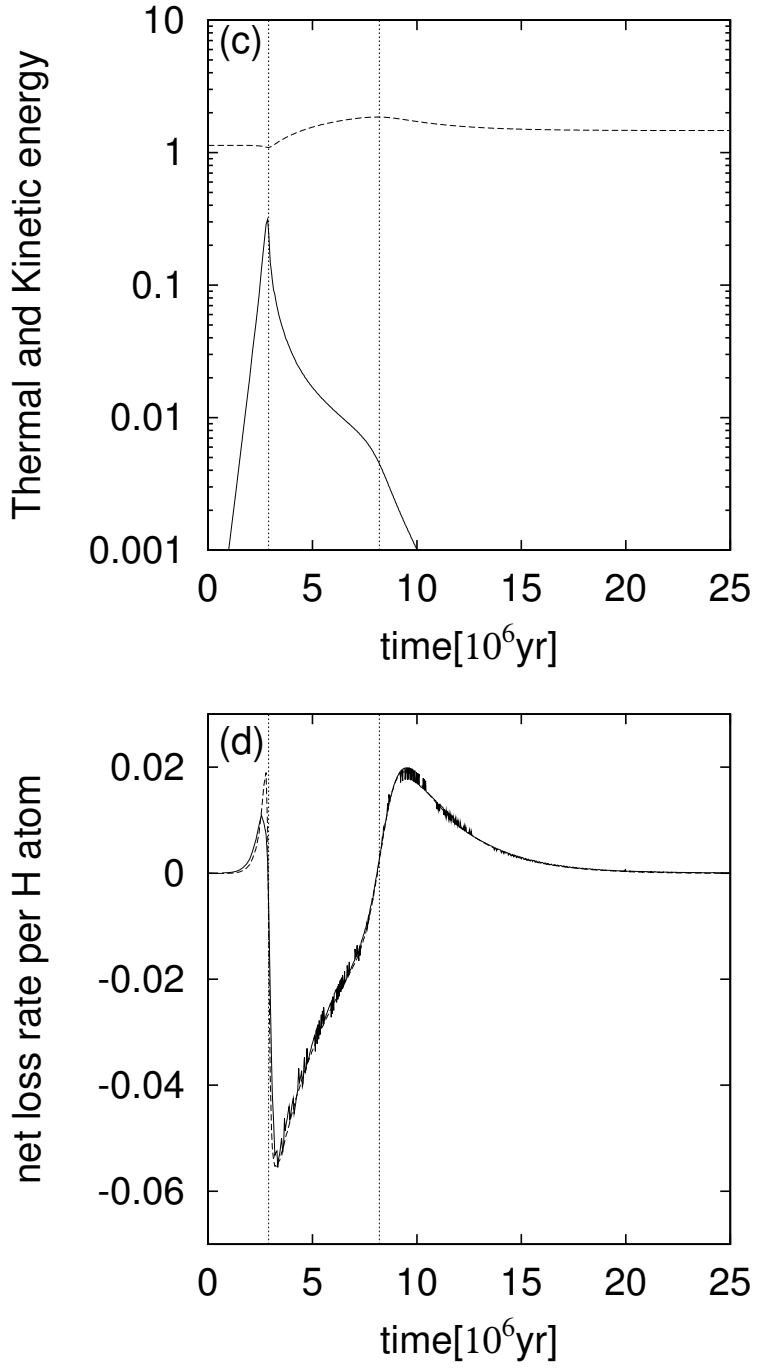


Figure 6.4: (a) Time evolution of the cold cloud (*solid line*) and intercloud (*dashed line*) density. (b) Time evolution of the maxmum (*solid line*) and minimum (*dashed line*) pressure. (c) Time evolution of the mean thermal energy per unit mass (*dashed line*). One thousand times the mean kinetic energy per unit mass is also shown (*solid line*). (d) Time evolution of the net radiative loss rete per H atom (*dashed line*). Solid line denotes $-\frac{1}{\gamma-1}\frac{\partial P}{\partial t}$.


(c)
Thermal and Kinetic energy
10
1
0.1
0.01
0.001
0
5
10
15
20
25
time[10^6yr]
(d)
net loss rate per H atom
0.02
0
-0.02
-0.04
-0.06
0
5
10
15
20
25
time[10^6yr]

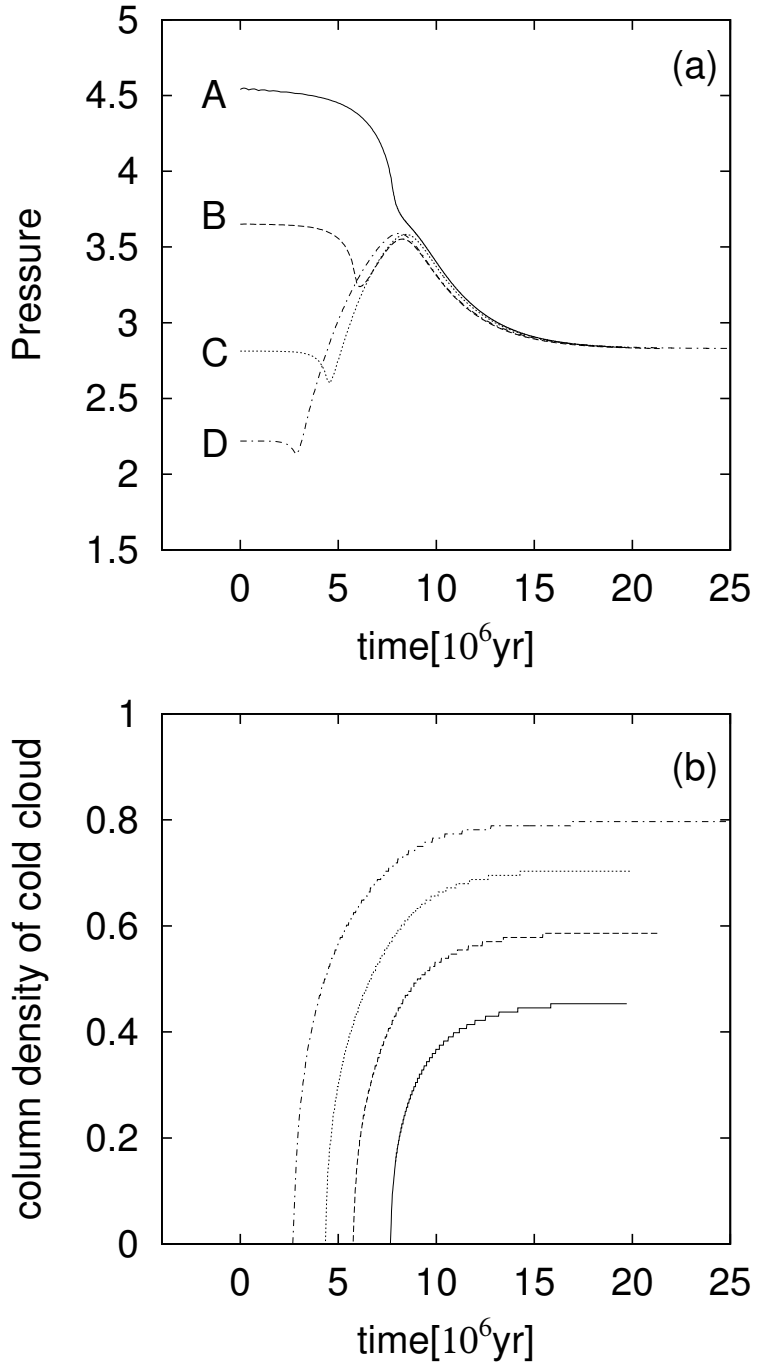
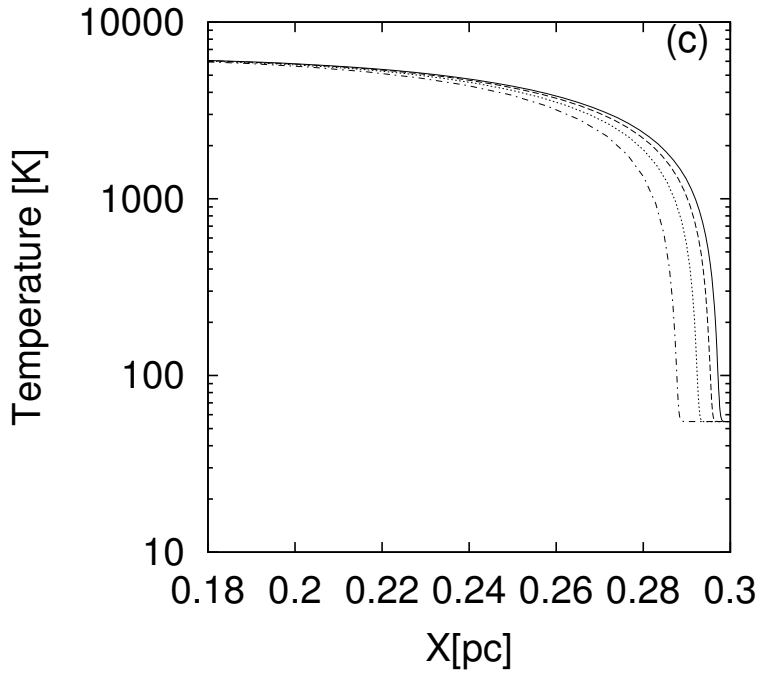
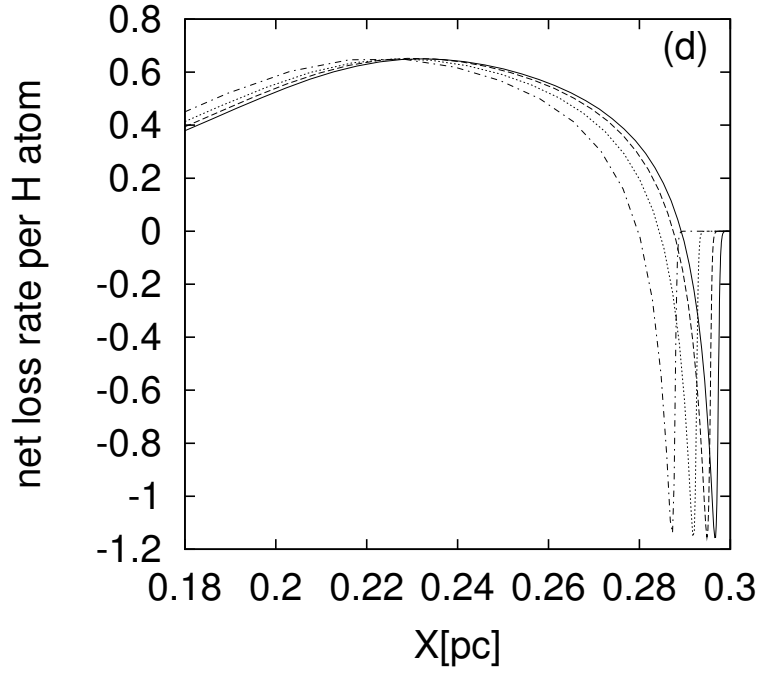


Figure 6.5: Various models with the same calculation domain (0.6 pc). (a) Time evolution of the pressure. (b) Column density of the forming cloud as a function of time. All grid elements with a density above 10 cm^{-3} are counted as belonging to the cold cloud. (c) Temperature structure through the cloud-intercloud interface. (d) the net loss rate per H atom.


(c)
10000
1000
100
10
Temperature [K]
0.18 0.2 0.22 0.24 0.26 0.28 0.3
X[pc]


(d)
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.2
net loss rate per H atom
0.18 0.2 0.22 0.24 0.26 0.28 0.3
X[pc]

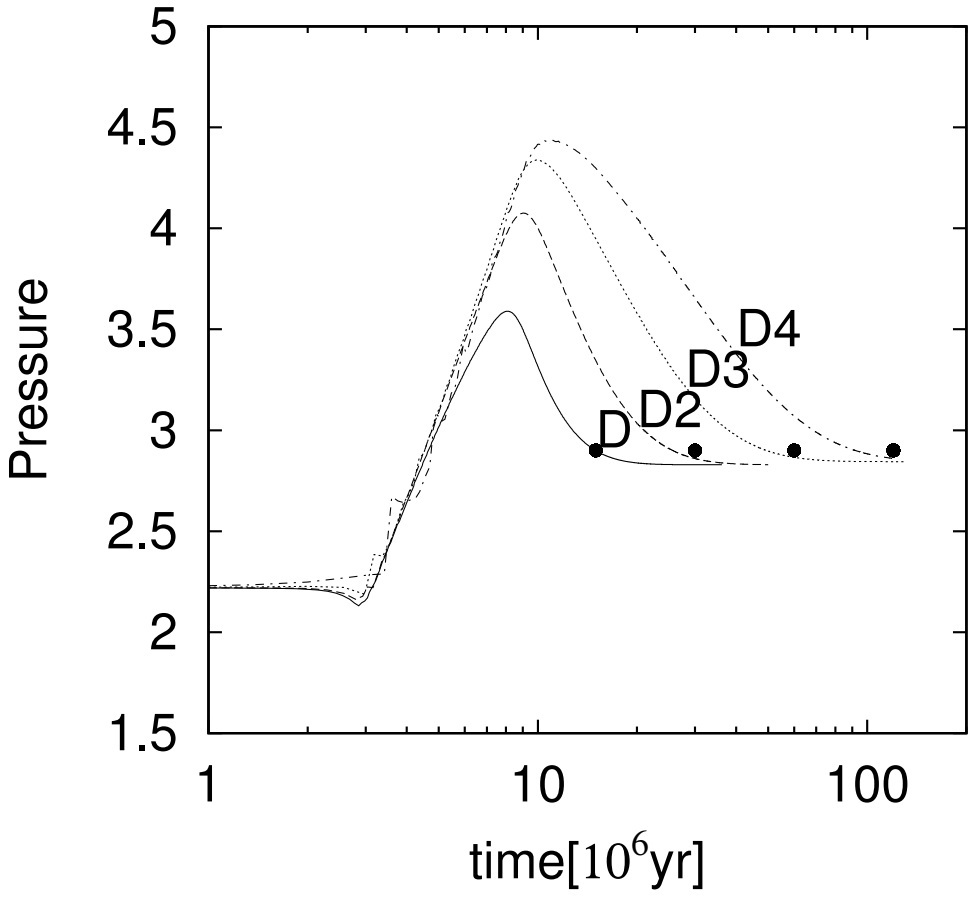


Figure 6.6: The calculations in the same initial parameter with various wavelength (model D: 0.6 pc, D2: 1.2 pc, D3: 2.4 pc, and D4: 4.8 pc). Filled circles denote the sound-crossing time $t = \alpha L/c_{\rm s}$ where $c_{\rm s}$ is the sound velocity of the warm medium.

7

SUMMARY AND CONCLUSIONS

We have examined the thermal and dynamical evolution of the interstellar medium using linear stability analysis and one and two-dimensional nonlinear calculations. The summary of this work is as follows.

1. In order to construct a theoretical model for interstellar medium, we have developed a numerical code for hydrodynamic calculations which includes all the necessary physical processes, such as photoelectric heating from small grains and PAHs, heating and ionization by cosmic rays and X-rays, heating by H_2 formation and destruction, atomic line cooling from hydrogen Lyα, C II, O I, Fe II, and Si II, rovibrational line cooling from H_2 and CO, and atomic and molecular collisions with grains.

2. We performed the linear stability analysis of thermal instability in isobarically contracting medium. As a result, isobarically uniform medium such as shock-compressed layers is thermally unstable in the temperature range of 300 – 7300 K. Our results of the linear analysis are used in the analysis of nonlinear calculation.

3. We have done one-dimensional hydrodynamic calculations for the propagation of a strong shock wave into a warm neutral medium (WNM) and a cold neutral medium (CNM), including detailed thermal and chemical processes. Our results show that the shock propagation into WNM can produce a thin and dense H_2 layer by the thermal instability inside the shock-compressed layer. The shock propagation into CNM can also produce the thermally collapsed layer. We predict that this thermally collapsed layer will fragment into small molecular cloudlets.

4. The propagation of a shock wave into an interstellar medium is investigated by two-dimensional numerical hydrodynamic calculation with cooling and thermal conduction. We present results from the first high resolution two-dimensional calculations ever to follow the fragmentation due to thermal instability in shock-compressed layer. We use realistic thermal conduction coefficient to resolve "Field length" that is the critical wavelength for thermal instability. We find that geometrically thin cooling layer behind the shock front fragments into small cloudlets. The cloudlets have supersonic velocity dispersion in the warm neutral medium in which the fragments are embedded as cold condensations. The fragments tend to coalesce and become larger clouds.

5. We investigate the evolution of interstellar two phase medium by one-dimensional hydrodynamic calculations with treatment of cooling and heating processes. Our main result shows that the true equilibrium of cold clouds and intercloud warm medium at the saturation pressure of the interstellar medium is attained within ~ 80 times the sound-crossing timescale in the intercloud warm medium, which is much shorter than the timescale previously believed.

The results of this thesis have revealed a very initial phase of cloud formation where cold cloudlets nucleate and coagulate into larger clouds. Although further studies on the effects of magnetic fields and possible scenarios for the formation of larger molecular clouds, etc. are necessary, we believe that the present work provides a foundation of the theory of molecular cloud formation.

APPENDIX

A TEST CALCULATIONS

A.1 Numerical Scheme

The hydrodynamics module of our scheme is based on the second-order Godunov method. We use the result of Riemann problem at each cell interface to calculate numerical fluxes, without introducing any artificial viscosity, which characterizes the Godunov method. Higher-order Godunov method has been the state-of-the-art numerical method in last two decades (see e.g., van Leer 1997), and widely used in astrophysical context (see e.g., Truelove et al. 1998).

We solve the hydrodynamic equations using the method of finite-differences with a time-explicit, multistep (operator split) solution procedure. An operator split method breaks the solution of partial differential equations into parts, with each part representing a single term in the equations. Each part is evaluated successively using the results from the update preceding it. Furthermore, there is considerable freedom in the choice of the order of the individual steps. However, numerical experiments have shown that such a multistep solutions procedure is more accurate than a single integration step based on old data, and that there is little difference between solutions generated using different orderings.

Our scheme is divided into four parts; Lagrangian step, conduction step, transport step, and cooling step. The transport step is used only on the Eularian grid calculations. We adopt the second-order accurate van Leer advection in the transport step (van Leer 1977). We adopt explicit time integration for thermal conduction term. We adopt linearized implicit time integration for the cooling and heating term.

A.2 Sod's Shock Tube Problem

In Figure A.1 and A.2, the results of 1D Sod's shock tube (Sod 1978) are plotted. Solid lines correspond to the analytic solution. The number of equal-mass grids in the x-direction is only 40 on the right-hand side of the initial discontinuity. Shock surfaces are resolved by small number of grids.

A.3 Blast Wave

Figure A.3 and A.3 plot the result of an extremely strong shock-tube problem (Woodward & Colella 1984) in which the Mach number was 200. We use 100 grids on each side of the

initial discontinuity. Initial pressure of the fluid on the left-hand side was 10^5 times that of right-hand side.

A.4 Thermal Conduction

Thermal conduction could stabilize temperature irregularites. Figure A.6 shows the results of test calculations for the thermal diffusion mode.

A.5 Growth rate at Linear phase

Finally, we investigate the growth rate at the early linear phase in order to confirm our numerical scheme. The numerical growth rates of the unstable and stable modes are shown in Figure A.7. We consider our numerical scheme has ability to simulate correctly the growth of perturbations due to the thermal instability.

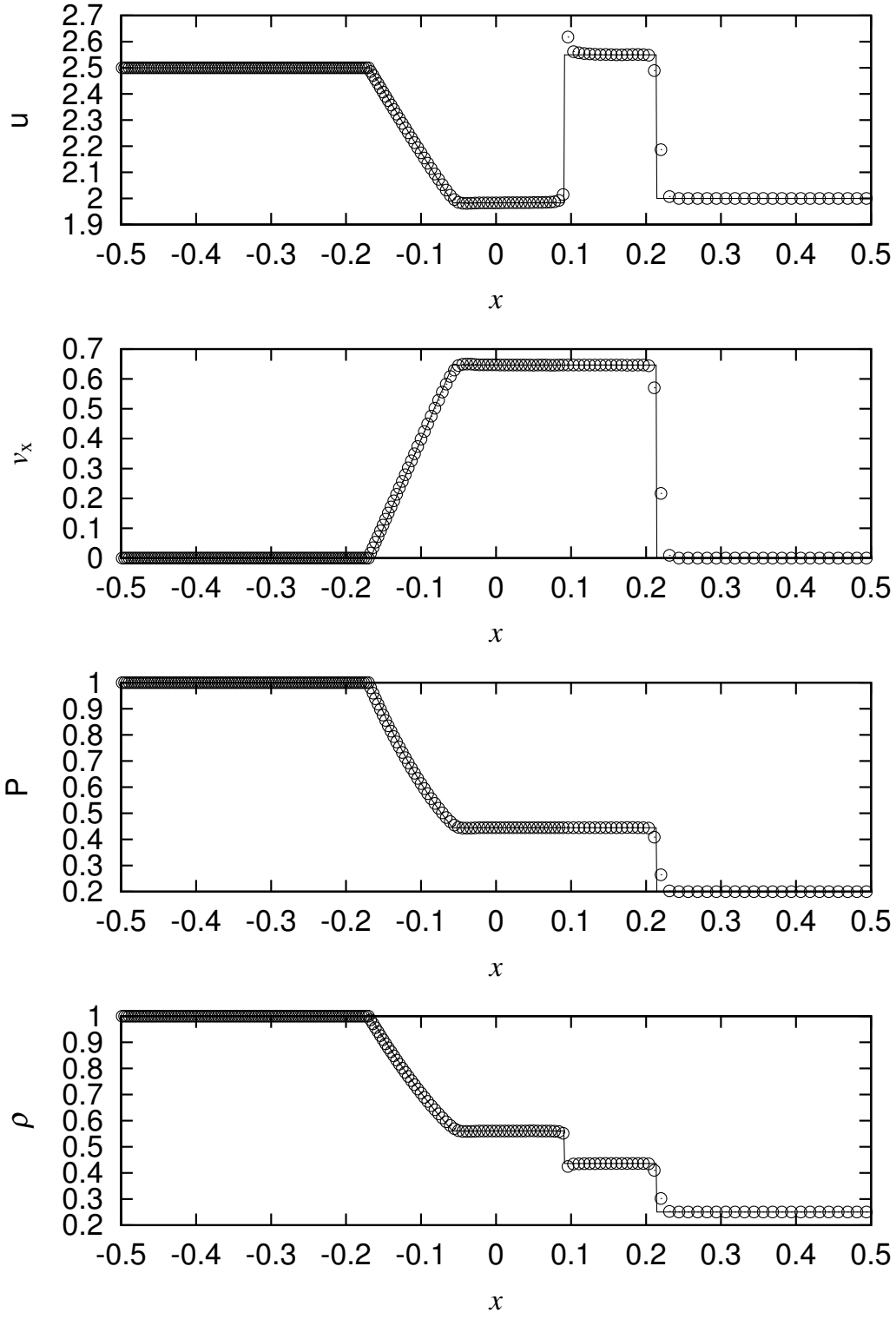


Figure A.1: Results of 1D calculation for the shock tube problem studied by Sod. The right hand side of the initial discontinuity includes 40 equal-mass Lagrangian grids.

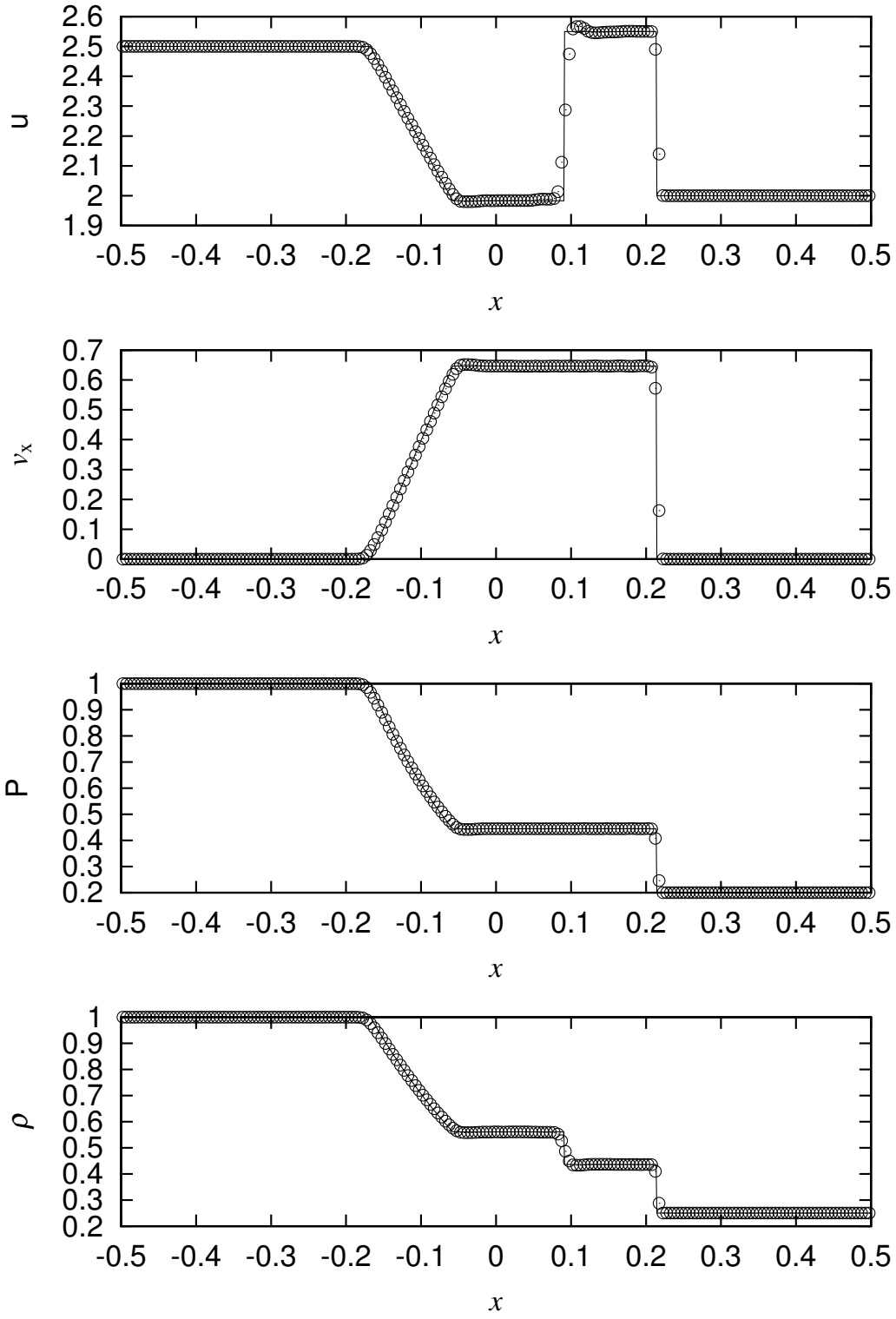


Figure A.2: Same as Figure A.1, but 200 uniformly separated Eulerian grid zones are used.

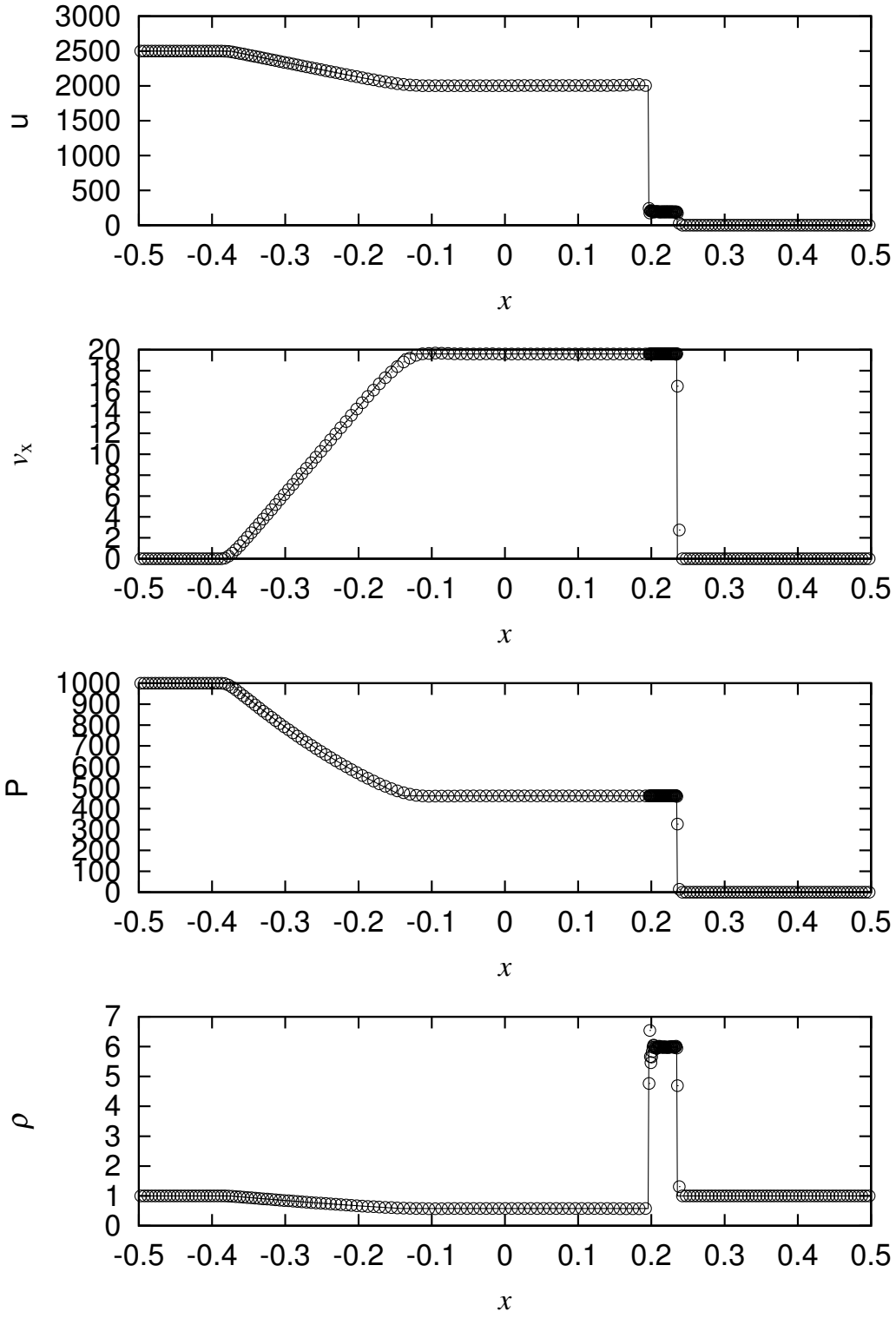


Figure A.3: Results of 1D calculation for the extremely strong shock tube problem of which Mach number is 200. Initially 100 equal-mass Lagrangian grids are distributed on both sides of the discontinuity.

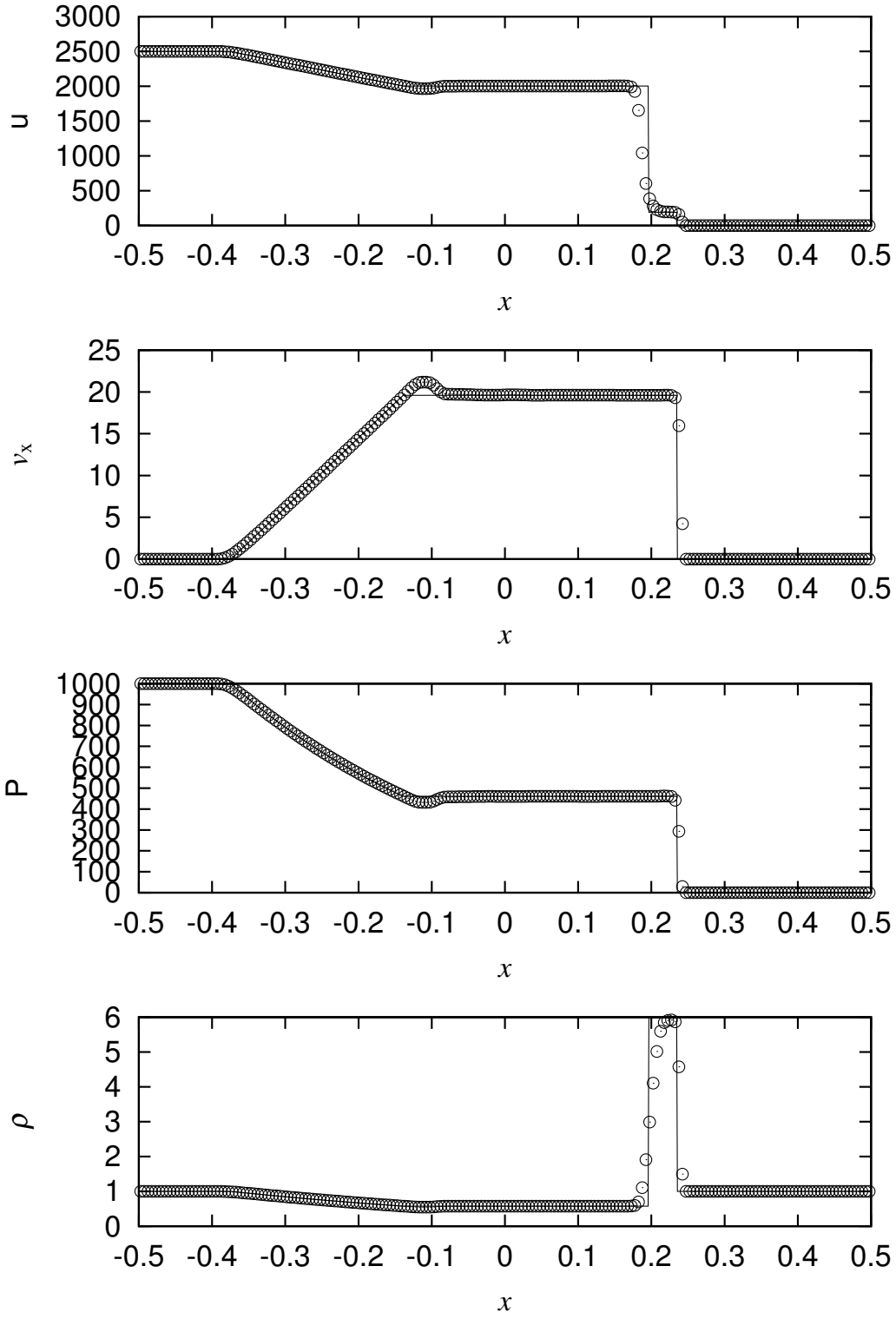


Figure A.4: Same as Figure A.3, but Eulerian scheme are used.

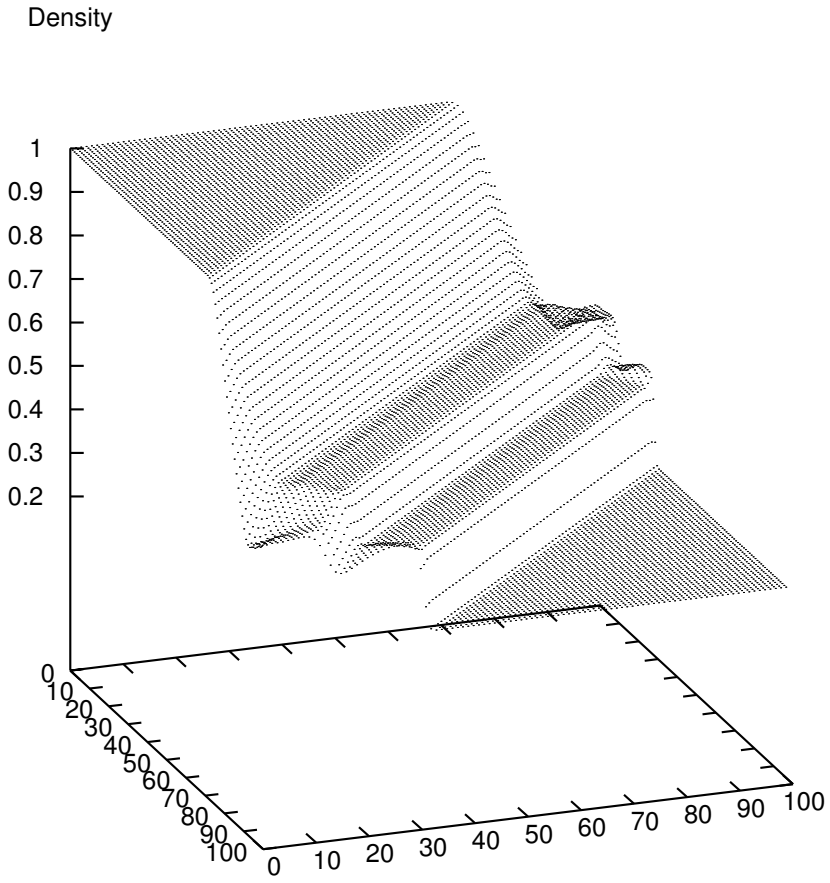


Figure A.5: Results of 2D calculation for the shock tube problem studied by Sod. 100 x 100 Cartesian grid points are used.

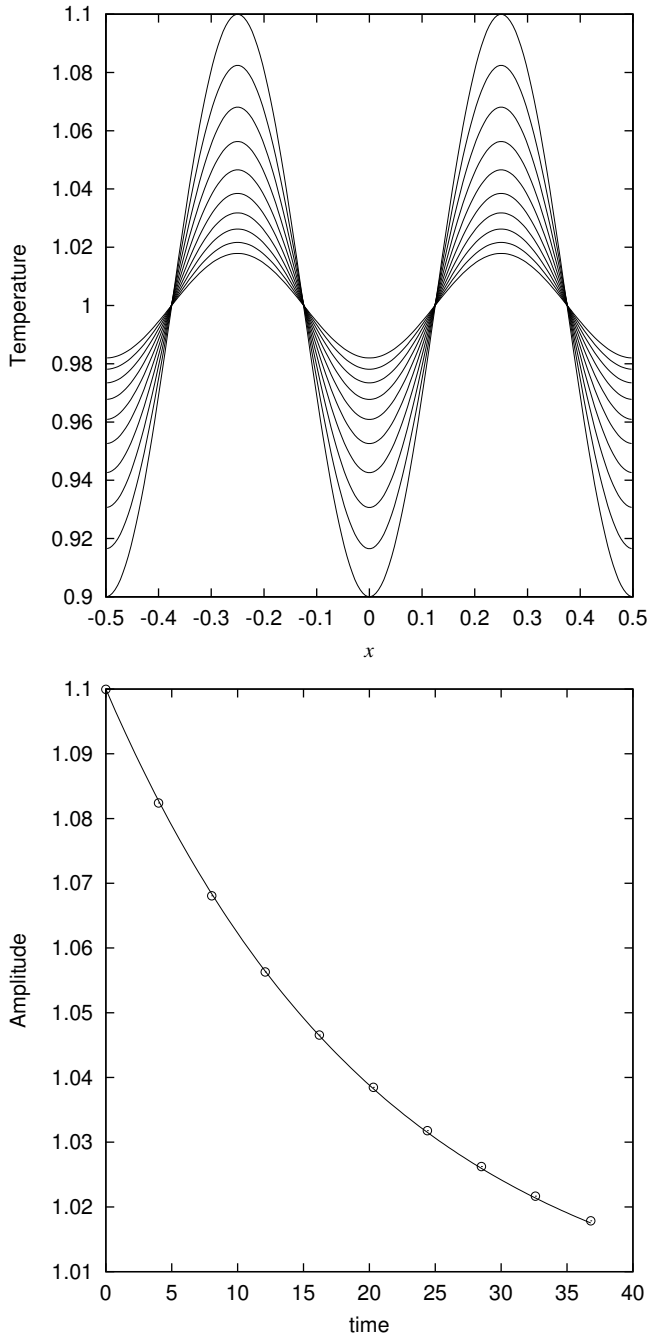


Figure A.6: Results of test calculations for the thermal diffusion mode.

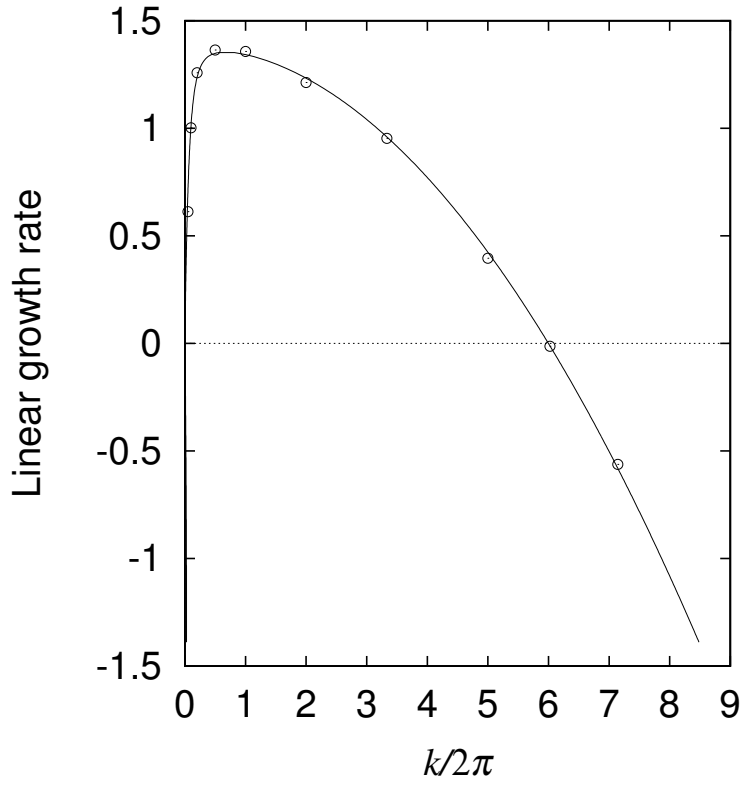


Figure A.7: Comparison between the growth rates obtained numerically and the analytic dispersion relation.

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